

TABLE 44.1 Selected Genetic Metabolic Disorders That Respond to Dietary Treatment—cont'd

Disorder	Affected Enzyme	Prevalence	Clinical and Biochemical Features	Medical Nutrition Therapy	Adjunct Treatment
Carbohydrate Disorders					
Galactosemia	Galactose-1-phosphate uridylyltransferase	1:48,000	Vomiting, hepatomegaly, FTT, cataracts, ID, often early sepsis ↑ urine and blood galactose	Eliminate lactose, low galactose, use soy protein isolate formula	
Hereditary fructose intolerance	Fructose-1-phosphate aldolase	1:20,000	Vomiting; hepatomegaly; hypoglycemia, FTT, renal tubular defects after fructose introduction ↑ blood and urine fructose after fructose feeding	No sucrose, fructose	
Fructose 1,6-diphosphatase deficiency	Fructose 1,6-diphosphatase	Unknown	Hypoglycemia, hepatomegaly, hypotonia, metabolic acidosis upon fructose introduction No ↑ blood/urine fructose	No sucrose, fructose	
Glycogen storage disease, type Ia	Glucose-6-phosphatase	1:100,000	Profound hypoglycemia, hepatomegaly	Low lactose, fructose, sucrose; low fat; high complex carbohydrate; avoid fasting	Raw cornstarch, iron supplements
Amino Acid Disorders					
Hyperphenylalaninemias					
Phenylketonuria	Phenylalanine hydroxylase	1:15,000		Food: low protein Formula: SF (without Phe, supplemented with tyrosine)	
Mild phenylketonuria	Phenylalanine hydroxylase	1:24,000	↑ blood Phe	Food: low protein Formula: SF (without Phe, supplemented with tyrosine)	
Dihydropteridine reductase deficiency	Dihydropteridine reductase	Rare	↑ blood Phe, irritability, developmental delay, seizures	Food: low protein Formula: SF (without Phe, supplemented with tyrosine)	Biopterin, 5-hydroxytryptophan, L-dopa, folic acid
Biopterin synthase defect	Biopterin synthase	Rare	Mild ↑ blood Phe, irritability, developmental delay, seizures	None	L-dopa, tetrahydrobiopterin, 5-hydroxytryptophan
Tyrosinemia, type I	Fumarylacetoacetate hydrolase	<1:100,000 to 1:120,000	Vomiting, acidosis, diarrhea, FTT, hepatomegaly, rickets ↑ blood and urine tyrosine, methionine; ↑ urine parahydroxy derivatives of tyrosine; liver cancer	Food: low protein Formula: SF (without tyrosine, Phe, methionine)	Nitisinone ^b
Maple Syrup Urine Disease (MSUD)					
MSUD	Branched-chain ketoacid decarboxylase complex (<2% activity)	1:185,000	Seizures, acidosis Plasma leucine, isoleucine, valine 10× normal	Food: low protein Formula: SF (without leucine, isoleucine, valine)	Thiamin ^c
Intermittent MSUD	Branched-chain ketoacid decarboxylase complex (<20% activity between episodes)	Rare	Intermittent symptoms Plasma leucine, isoleucine, valine 10× normal during illness	Food: low protein Formula: SF (without leucine, isoleucine, valine)	

Continued

TABLE 44.1 Selected Genetic Metabolic Disorders That Respond to Dietary Treatment—cont'd

Disorder	Affected Enzyme	Prevalence	Clinical and Biochemical Features	Medical Nutrition Therapy	Adjunct Treatment
Homocystinuria	Cystathionine synthase or similar	1:200,000	Detached retinas; thromboembolic and cardiac disease; mild to moderate ID; bony abnormalities; fair hair and skin; ↑ methionine, homocysteine	Food: low protein Formula: SF (without methionine, supplemented with L-cystine)	Betaine, folate, vitamin B ₁₂ , vitamin B ₆ ^c if folate levels are normal
Fatty Acid Oxidation Disorders					
Long-chain acyl-CoA dehydrogenase deficiency	Long-chain acyl-CoA dehydrogenase	Rare	Vomiting, lethargy, hypoglycemia	Low fat, low long-chain fatty acids; avoid fasting	MCT oil, L-carnitine ^d
Long-chain 3-hydroxy-acyl-CoA dehydrogenase deficiency	Long-chain 3-hydroxy-acyl-CoA dehydrogenase	Rare	Vomiting, lethargy, hypoglycemia	Low fat, low long-chain fatty acids; avoid fasting	MCT oil, L-carnitine [§]
Medium-chain acyl-CoA dehydrogenase deficiency	Medium-chain acyl-CoA dehydrogenase	1:13,000 to 1:19,000	Vomiting, lethargy, hypoglycemia	Low fat, low medium-chain fatty acids, avoid fasting	L-carnitine ^d
Short-chain acyl-CoA dehydrogenase deficiency	Short-chain acyl-CoA dehydrogenase	1:35,000	Vomiting, lethargy, hypoglycemia	Low fat, low short-chain fatty acids, avoid fasting	L-carnitine ^d
Very-long-chain acyl-CoA dehydrogenase deficiency	Very-long-chain acyl-CoA dehydrogenase	1:30,000 to 1:100,000	Vomiting, lethargy, hypoglycemia	Low fat, low long-chain fatty acids, avoid fasting	L-carnitine, ^d MCT oil

^aPhenylbutyrate is a chemical administered to enhance waste ammonia excretion; other compounds producing the same effect are also used.

^bNitisinone, formerly NTBC, 2-(2-nitro-4-trifluoromethyl-benzoyl-1,3-cyclohexanedione), commercially available as Orfadin and Nityr.

^cPatient may or may not respond to the compound.

^dUse depends on clinic.

CoA, Coenzyme A; FTT, failure to thrive; ID, intellectual disability; IV, intravenous; MCT, medium-chain triglyceride; MSUD, maple syrup urine disease; Phe, phenylalanine; SF, specialized formulas are available for medical nutrition therapy for this disorder; UCD, urea cycle disorder.

FOCUS ON

Newborn Screening

Since the 1960s, states across the United States have adopted mandatory newborn screening (NBS) as law (Waisbren, 2006). These programs were developed as a result of the efficacy of the Guthrie bacterial inhibition assay, in which dried blood spots were used to identify phenylketonuria (PKU). This simple, sensitive, and inexpensive screening test became the basis for population-based screening systems for newborns. Hemoglobinopathies, endocrine disorders, metabolic disorders, and some infectious diseases can be identified effectively with the use of dried blood spots.

Tandem mass spectrometry was first used in the 1990s and is now used across the United States. This technology makes it possible to identify multiple disorders from a single dried blood spot. The number of disorders screened for varies by state, and expanded screening also is offered by private, for-profit companies. Follow-up programs also vary; some states have single, organized programs, whereas follow-up in other states is less centralized. Conditions successfully screened for by early NBS programs include congenital hypothyroidism, PKU, congenital adrenal hyperplasia, galactosemia, sickle cell disease, and maple syrup urine disease (Brosco et al, 2006).

The Maternal and Child Health Bureau (Maternal Child Health Bureau (MCHB), 2006) of the U.S. Health Resources and Services Administration commissioned a report from the American College of Medical Genetics (ACMG). This expert panel identified 29 conditions for which newborn screening should be mandated

and 25 secondary conditions that may be detected incidentally (Watson et al., 2006). The ACMG developed a series of ACTION (ACT) sheets and confirmatory algorithms for disorders that are identified by NBS screening. The ACT sheets describe the steps health professionals should follow in communicating with the family and determining follow-up (ACMG, 2001).

Other groups, including the World Health Organization, March of Dimes, and Massachusetts Newborn Screening Advisory Committee, also have issued recommendations.

Providers who may be involved in the care and follow-up of families identified by NBS should have a good understanding of their state's system, as well as the confounding factors that may affect results. Communication among families, primary health care providers, and tertiary clinics is critical to timely identification and treatment. Follow-up, including referral to the appropriate specialists, is important for any family who receives positive NBS results. NBS fact sheets from the Committee on Genetics of the American Academy of Pediatrics describe (1) newborn screening; (2) follow-up of abnormal screening results to facilitate timely diagnostic testing and management; (3) diagnostic testing; (4) disease management, which requires coordination with the medical home and genetic counseling; and (5) continuous evaluation and improvement of the NBS system (Kaye et al, 2006).



Fig. 44.1 Blood spots are collected from a newborn for newborn screening. (Courtesy Kelly McKean.)

Requirements for individual amino acids are difficult to determine because typical growth and development can be achieved over a wide range of intake. The data of Holt and Snyderman (1967) often are used as the basis for prescribing amino acid intakes (Table 44.2). Careful and frequent monitoring is required to ensure the adequacy of the nutritional prescription. Although nitrogen studies are the most precise, weight gain in infants is a sensitive and easily monitored index of well-being and nutritional adequacy.

PHENYLKETONURIA

Etiology

Phenylketonuria (PKU) is the most common of the hyperphenylalaninemias. In this disorder, Phe is not metabolized to Tyr because of a deficiency or inactivity of phenylalanine hydroxylase (PAH), as shown in Fig. 44.2. Of the amino acid disorders, PKU provides a reasonable model for detailed discussion because it (1) occurs relatively frequently and most neonates are screened for it; (2) has a successful MNT treatment; and (3) has a predictable course, with available documentation of “natural” and “intervention” history (see *Focus On: Timeline of Events in the Diagnosis and Treatment of Phenylketonuria*).

Nutritional treatment involves restricting the substrate (Phe) and supplementing the product (Tyr) (see *Pathophysiology and Care Management Algorithm: Phenylketonuria*). Most affected infants exhibit PAH deficiency; the remainder (<3%) have defects in associated pathways. Low-Phe nutrition therapy does not prevent the neurologic deterioration present in the disorders of these other associated pathways.

Medical Treatment

All states have newborn screening programs for PKU and other metabolic disorders. Diagnostic criteria for PKU include an elevated blood concentration of Phe and an elevated (i.e., >3) Phe:Tyr ratio. The diagnostic process also should include evaluation for hyperphenylalaninemia that results from the deficiency of enzymes other than PAH, including defects in tetrahydrobiopterin (BH₄) synthesis or regeneration (Vockley et al, 2014). An effective newborn screening program and access to an organized follow-up program are critical to the early identification and treatment of infants with PKU.

TABLE 44.2 Approximate Daily Requirements for Selected Dietary Components and Amino Acids in Infancy and Childhood

Dietary Component or Amino Acid	AGE AND REQUIREMENT	
	Birth to 12 Months (mg/kg)	1–10 Years (mg/day)
Phenylalanine	1–5 months: 47–90 6–12 months: 25–47	200–500 ^a
Histidine	16–34	
Tyrosine ^b	1–5 months: 60–80 6–12 months: 40–60	25–85 (mg/kg)
Leucine	76–150	1000
Isoleucine	1–5 months: 79–110 6–12 months: 50–75	1000
Valine	1–5 months: 65–105 6–12 months: 50–80	400–600
Methionine ^c	20–45	400–800

Dietary Component or Amino Acid	AGE AND REQUIREMENT	
	Birth to 12 Months (mg/kg)	1–10 Years (mg/day)
Cyst(e)ine ^d	15–50	400–800
Lysine	90–120	1200–1600
Threonine	45–87	800–1000
Tryptophan	13–22	60–120
Energy	1–5 months: 108 kcal/kg 6–12 months: 98 kcal/kg	70–102 kcal/kg
Water	100 mL/kg	1000 mL
Carbohydrate	kcal $\times$ 0.5 $\div$ 4 = g/day	kcal $\times$ 0.5 $\div$ 4 = g/day
Total protein	1–5 months: 2.2 g/kg 6–12 months: 1.6 g/kg	16–18
Fat	kcal $\times$ 0.35 $\div$ 9 = g/day	kcal $\times$ 0.35 $\div$ 9 = g/day

^aMore phenylalanine (>800 mg) is required in the absence of tyrosine.

^bTotal phenylalanine plus tyrosine should be considered in the prescription because most phenylalanine is converted to tyrosine.

^cMore methionine is required in the absence of cyst(e)ine.

^dMore cyst(e)ine is required in presence of a blocked *trans*-sulfuration outflow pathway for methionine metabolism.

Compiled from amino acid data of Holt and Snyderman. Information on amino acid requirements of infants and children at different ages is limited; the figures given here are in excess of minimum requirements. Consequently, this table should be used only as a guide and should not be regarded as an authoritative statement to which individual patients must conform.

(Modified from American Academy of Pediatrics, Committee on Nutrition: Special diets for infants with inborn errors of metabolism, *Pediatrics* 57:783, 1976.)

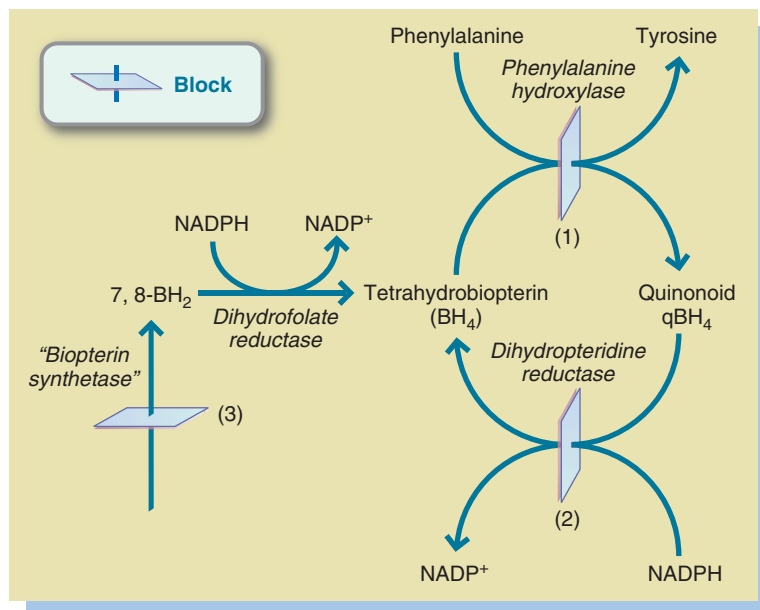


Fig. 44.2 Hyperphenylalaninemias. (1) Phenylalanine hydroxylase deficiency; (2) Dihydropteridine reductase deficiency; (3) Biopterin synthetase deficiency. *NADPH*, Nicotinamide-adenine dinucleotide phosphate (reduced form); *NADP⁺*, nicotinamide-adenine dinucleotide phosphate (oxidized form).

FOCUS ON

Timeline of Events in the Diagnosis and Treatment of Phenylketonuria

1934: A. Folling identifies phenylpyruvic acid in the urine of mentally retarded siblings.

1950s: G. Jervis demonstrates a deficiency of phenylalanine oxidation in the liver tissue of an affected patient. H. Bickel demonstrates that dietary phenylalanine restrictions lower the blood concentration of phenylalanine.

1960s: R. Guthrie develops a bacterial inhibition assay for measuring blood phenylalanine levels.

Mid-1960s: Semisynthetic formulas restricted in phenylalanine content become commercially available.

1965–1970: States adopt newborn screening programs to detect phenylketonuria (PKU).

1967–1980: Collaborative Study of Children Treated for Phenylketonuria is conducted. Data from this study form the basis for treatment protocols for PKU clinics in the United States.

Late-1970s: Detrimental effects of maternal PKU are recognized as a significant public health problem.

1980s: Lifelong restriction of phenylalanine intake becomes the standard of care in PKU clinics in the United States.

1983: The Maternal PKU Collaborative Study begins to study the effects of treatment on the pregnancy outcome of women with phenylketonuria.

1987: Techniques for carrier detection and prenatal diagnosis of PKU are developed.

Late-1980s: The gene for phenylalanine hydroxylase deficiency (MIM No. 261600) is located on chromosome 12q22-q24.1. DNA mutation analysis is accomplished with peripheral leukocytes.

1990s: Phenylalanine level of 2–6 mg/dL (120–360 μmol/L), lower than the previous level of less than 10 mg/dL (600 μmol/L), becomes the new standard of care for treatment of PKU.

2000s: Tetrahydrobiopterin-responsive forms of PKU are recognized, especially those with mild mutations.

2007: Sapropterin dihydrochloride (Kuvan), the commercial form of tetrahydrobiopterin, receives FDA approval.

2010: Research into alternative and adjunct therapies such as the use of large neutral amino acids, enzyme substitution, and somatic gene therapy continues.

2014: Clinical trials for enzyme replacement therapy using pegylated phenylalanine ammonia lyase (PEG-PAL) injections are underway.

2018: Enzyme replacement therapy (pegvaliase-pqpz injection) receives FDA approval to lower blood-Phe levels in adults who have uncontrolled Phe levels on current treatment.

(Data from Maternal Child Health Bureau: Newborn screening: toward a uniform screening panel and system, *Genet Med* 8(Suppl 1):1 S, 2006; Saugstad LF: From genetics to epigenetics, *Nutr Health* 18:285, 2006; Mitchell JJ, Scriver CR: Phenylalanine hydroxylase deficiency. In Pagon RA, et al, editors: *GeneReviews* [website]. Seattle, 1993–2000, University of Washington [updated January 5, 2017].)

The advantage of rigorous nutrition therapy has been demonstrated by measurements of intellectual function. Individuals who do not receive diet therapy have severe intellectual disability, whereas individuals who are on therapy from the early neonatal period have normal intellectual function (McPheeters et al, 2012). Outcome, measured as intellectual function, depends on the age of the infant at diagnosis and

start of nutrition therapy, as well as the individual's biochemical control over time.

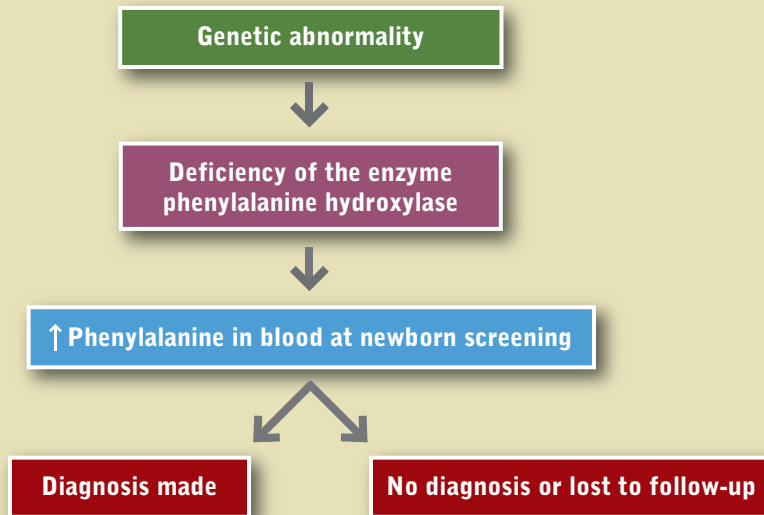
BH₄ has been studied to evaluate its effectiveness as an alternative treatment to severe dietary Phe restriction since BH₄ is a cofactor needed for proper activity of the enzyme. Treatment with sapropterin (a synthetic form of BH₄) is used as an adjunct therapy for some,



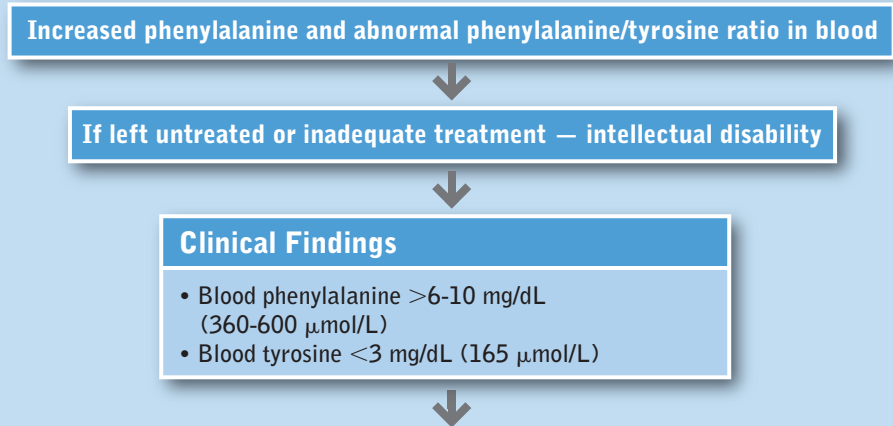
PATHOPHYSIOLOGY AND CARE MANAGEMENT ALGORITHM

Phenylketonuria

ETIOLOGY



PATHOPHYSIOLOGY



MANAGEMENT

Medical Management

Regular monitoring of blood phenylalanine to maintain at 1-6 mg/dL (60-360 $\mu\text{mol/L}$)

Nutrition Care

- Phenylalanine-free formula/medical food
- Low phenylalanine foods
- Supplement with tyrosine
- Education of family and child about formula/medical food preparation
- Adequate nutritional intake
- Regular monitoring of growth
- Education on label reading and food choices

especially with milder mutations (Longo et al, 2015). Those individuals who respond have what is called BH_4 -responsive PKU. However, even for BH_4 -responsive individuals, ongoing intervention and nutrition monitoring are needed (Singh et al, 2010).

Enzyme replacement with phenylalanine ammonia lyase (PAL) was approved for use in adults with high blood-Phe levels in 2018. This

medication (pegvaliase-pqpz or Palynziq) is administered through daily injections. Gene therapy to restore PAH activity is also being studied.

Blood phenylalanine control. Blood-Phe concentration must be checked regularly, depending on the age and health status of the child, to be sure it remains within the range of 2 to 6 mg/dL or 120 to

360 $\mu\text{mol/L}$ (McPheeters et al, 2012). Phe-containing foods are offered as tolerated as long as the blood concentration of Phe remains in the range of good biochemical control. The child's rate of growth and mental development must be monitored carefully.

Effective management requires a team approach in which the child, parents, registered dietitian nutritionist, pediatrician, psychologist, social worker, and nurse work together to achieve and maintain biochemical control in an atmosphere promoting normal mental and emotional development. An essential management tool for parents, children, and clinicians is the food diary used to monitor Phe intake. Daily record keeping supports compliance with treatment and builds self-management skills. An accurate record of food and formula intake for at least the 3 days before a laboratory specimen is obtained is critical for accurate interpretation of the results and subsequent adjustment of the Phe prescription.

Elevations in blood-Phe concentration generally are caused by either excessive Phe intake or tissue catabolism. Intake of Phe in excess of the amount required for growth accumulates in the blood. Deficient energy intake or the stress of illness or infection can result in protein breakdown and the release of amino acids, including Phe, into the blood. In general, the anorexia of illness limits energy intake. Preventing tissue catabolism by maintaining intake of the formula/medical food as much as possible is essential. Although it occasionally may be necessary to offer only clear liquids during an illness, the Phe-free formula/medical food should be reintroduced as soon as it is feasible. Tube-feeding is an option if oral intake is not possible.

Continuing the restricted-Phe dietary therapy throughout childhood, adolescence, and beyond is recommended (McPheeters et al, 2012; Vockley et al, 2014). Progressively decreasing IQs, learning difficulties, poor attention span, and behavioral difficulties have been reported in children who have discontinued the dietary regimen. Children who maintain well-controlled blood-Phe levels demonstrate comparatively higher intellectual achievement than those who do not. Good dietary control of blood-Phe concentrations is the best predictor of IQ, whereas "off-diet" blood-Phe concentrations of greater than 20 mg/dL (1200 $\mu\text{mol/L}$) are the best predictors of IQ loss. Subtle deficits in higher-level cognitive function may persist even at blood-Phe levels of 6 to 10 mg/dL (360 to 600 $\mu\text{mol/L}$). Thus, most clinics recommend treatment blood levels of 2 to 6 mg/dL (120 to 360 $\mu\text{mol/L}$). Restricted-Phe therapy should be continued for life to maintain normal cognitive function.

Medical Nutrition Therapy

Nutrition management guidelines for PKU have been published (Singh et al, 2014).

Formula. For PKU, dietary therapy is planned around the use of a formula/medical food with Phe removed from the protein. Formulas or medical foods provide a major portion of the daily protein and energy needs for affected infants, children, and adults. In general, the protein source in the formula/medical food is L-amino acids with the critical amino acid (i.e., Phe) omitted. Glycomacropeptide (GMP), a whey protein with very little Phe, is used in a few medical foods as an alternative to L-amino acids (van Calcar and Ney, 2012). Carbohydrate sources are corn syrup solids, modified tapioca starch, sucrose, and hydrolyzed cornstarch. Fat is provided by a variety of oils.

Some formula/medical foods contain no fat or carbohydrate, so these components must be provided from other sources. If using formulas without fat, clinicians must provide sources of essential fatty acids. Essential fatty acid deficiencies have been noted among individuals consuming fat-free formulas (Camp et al, 2014; Singh et al, 2014). Most formulas and medical foods contain calcium, iron, and all other necessary vitamins and minerals and are a reliable source of these

nutrients. When others are devoid of these nutrients, supplementation is needed to ensure nutritional adequacy.

Phe-free formula is supplemented with regular infant formula or breastmilk during infancy and cow's milk in early childhood to provide high-biologic value protein, nonessential amino acids, and sufficient Phe to meet the individualized requirements of the growing child. The optimal amount of protein substitute depends on the individual's age (and thus requirements for growth) and enzyme activity; thus, it must be prescribed individually. Because the protein in specialized formulas is synthetic, it is provided in amounts greater than the dietary reference intake (DRI) (Singh et al, 2014).

The Phe-free formula and milk mixture often provides approximately 90% of the protein and 80% of the energy needed by infants and toddlers. A method for calculating the appropriate quantities of a low-Phe food pattern is shown in Table 44.3. Calculations should provide adequate but not excessive energy for the infant, as well as appropriate fluid to maintain hydration. To support metabolic control effectively, formula/medical foods must be consumed in three or four nearly equal portions throughout the day.

Low-phenylalanine foods. Foods of moderate- or low-Phe content are used as a supplement to the formula or medical food mixture. These foods are offered at the appropriate ages to support developmental readiness and to meet energy needs. Puréed foods from a spoon may be introduced at 5 to 6 months of age, finger foods at 7 to 8 months, and the cup at 8 to 9 months, using the same timing and progression of texture recommended for typical children (see Chapters 15 and 16). Table 44.4 lists typical low-Phe food patterns for young children.

Low-protein pastas, breads, and baked goods made from wheat starch add variety to the food pattern and allow children to eat some foods "to appetite." A variety of low-protein pastas, rice, baked goods, egg replacers, and other foods are available. Wheat starch and a variety of low-protein baking mixes for breads, cakes, and cookies are also available. Table 4.5 compares low-protein and regular food items.

In many cases, parents create recipes or adapt family favorites to meet the needs of their children. These recipes offer the children a variety of textures and food choices, allowing them to participate in family meals. Families are also able to meet the energy and Phe needs of their children without resorting to excessive intakes of sugars and concentrated sweets.

A formula or medical food that is free of Phe and has a more appropriate amino acid, vitamin, and mineral composition for an older child is generally introduced in the toddler or preschool period. The criteria for the introduction of these "next-step" formulas are that the child accepts the food pattern and formula well and reliably consumes a wide variety of foods from the low-Phe food list. Successful management with consistently low blood-Phe levels is based on habit (i.e., the formula/medical food is offered and consumed without negotiation or threat). Children respond favorably to the regularity of the time of ingestion of the formula/medical food and the familiarity of its taste and presentation. Table 44.6 compares a restricted-Phe food pattern with a typical food pattern for a child of the same age.

Education about therapy management. The energy needs and amino acid requirements of children with PKU do not differ appreciably from those of children in general. With proper management, typical growth can be expected (Fig. 44.3). However, parents may tend to offer excessive energy as sweets because they feel their child is being deprived of food experiences. Health care providers should support families in recognizing that children with PKU are healthy children who must make careful food choices for themselves, not chronically ill children who require food indulgences.

Appropriate clinical interaction with family members provides them with the information and skills to differentiate between food

TABLE 44.3 Guidelines for Low-Phenylalanine Food Pattern Calculations

Step 1: Calculate the child's needs for phenylalanine, protein, and energy (kcal).

Phenylalanine

$$7.7 \text{ kg} \times 40^a \text{ mg phenylalanine} = 308 \text{ mg Phe/kg/day}$$

Protein

$$7.7 \text{ kg} \times 2.5\text{--}3^b \text{ g pro} = 19\text{--}25 \text{ g pro/kg/day}$$

Energy

$$7.7 \text{ kg} \times 100^c \text{ kcal} = 770 \text{ kcal/kg/day}$$

Step 2: Estimate the amount of phenylalanine, protein, and energy to be obtained from foods other than the formula mixture.

APPROXIMATE NEEDS

	Phenylalanine (mg/day)	Protein (g/day)	Energy (kcal/day)
Foods	30	1–2	100
Formula	~288	18–24	670
<i>Total</i>	<i>318</i>	<i>19–25</i>	<i>770</i>

Step 3: Confirm that a reasonable food pattern is provided with the estimated phenylalanine from food guideline. A sample food pattern:

	Phenylalanine (mg)	Protein (g)	Energy (kcal)
Green beans, strained, 2 Tbsp	15	0.4	8
Banana, mashed, 2 Tbsp	10	0.3	28
Carrots, strained, 2 Tbsp	6	0.2	7
<i>Total</i>	<i>31</i>	<i>0.9</i>	<i>43</i>

Step 4: Determine the amount of standard infant formula to be included in the formula. This information is determined from the infant's estimated phenylalanine needs.

Step 5: Determine the amount of phenylalanine-free formula required per day. This information is determined from the infant's estimated protein and energy needs.

Step 6: Determine the amount of water to mix with the formula. The consistency of the formula will vary according to the infant's age and fluid requirements. For example, for the infant described in the case study, add water to make a total volume of 32 oz.

Step 7: Determine the amounts of phenylalanine, protein, and energy provided by formulas and foods.

	Phenylalanine (mg/day)	Protein (g/day)	Energy (kcal/day)
Phenex-1 powder (70 g)	0	10.5	325
Enfamil powder (70 g)	294	7.4	350
Food	30	1–2	100
<i>Total</i>	<i>324</i>	<i>18.9–19.9</i>	<i>775</i>

Step 8: Determine the actual amounts of phenylalanine, protein, and energy per kilogram body weight.

Phenylalanine

$$324 \div 7.7 \text{ kg} = 42 \text{ mg Phe/kg/day}$$

Protein

$$19.4 \div 7.7 \text{ kg} = 2.5 \text{ g pro/kg/day}$$

Energy

$$774 \div 7.7 \text{ kg} = 101 \text{ kcal/kg/day}$$

^aA phenylalanine intake of 40 mg/kg/day is chosen as a moderate intake level. The prescription for phenylalanine must be adapted to individual needs as judged by growth and blood levels.

^bAlthough these intakes are higher than the recommended dietary allowance, they are the intakes found by the Collaborative Study to promote normal growth with consumption of protein hydrolysate-based formula.

^cTotal energy intake must be adjusted to meet individual needs, and an excess must be avoided.

(From Acosta PB: Recommendations for protein and energy intakes by patients with phenylketonuria. *Eur J Pediatr* 155(Suppl 1):S121, 1996; Singh RH, Rohr F, Frazier D, et al: Recommendations for the nutrition management of phenylalanine hydroxylase deficiency. *Genet Med* 16:121, 2014.)

behaviors that are typical for the age and developmental level of the child and those related specifically to PKU. To avoid power struggles and conflicts over food, it is advisable to involve the child in choosing appropriate foods at an early age. Children who are 2 to 3 years old can master the concept of appropriate choices when foods are categorized as “yes foods” and “no foods.” The concept of an appropriate quantity of a food can be introduced to a 3- or 4-year-old child in terms of “how many” by counting crackers or raisins and then in terms of “how much”

by weighing or measuring foods such as cereal or fruit. The child then moves to more complex tasks (e.g., formula and food preparation) and planning of meals (e.g., breakfast or a packed lunch). Responsibility for planning a full day's menu by calculating the quantity of Phe in portions of food and compiling the daily total is the ultimate goal. These age-related tasks are shown in [Table 44.7](#).

Psychosocial development. The necessity of carefully controlling food intake may prompt parents to overprotect their children and

TABLE 44.4 Typical Menus for a 3-Year-Old with Phenylketonuria

Tolerance: 300 mg phenylalanine/day		Tolerance: 400 mg phenylalanine/day	
Formula/medical food for 24 h: 120 g Phe-free-2, 100 g 2% milk, water to 24 oz		Formula/medical food for 24 h: 120 g Phe-free-2, 100 g 2% milk, water to 24 oz	
This formula mixture provides 30 g protein, 460 kcal, 170 mg phenylalanine.		This formula mixture provides 30 g protein, 460 kcal, 170 mg phenylalanine.	
Menu for 100 mg Phenylalanine from Food	Phenylalanine (mg)	Menu for 200 mg Phenylalanine from Food	Phenylalanine (mg)
Breakfast		Breakfast	
Formula mixture, 6 oz		Formula mixture, 6 oz	
Kix cereal, (¼ cup)	22	Rice Krispies, 20 g (¼ cup)	25
Peaches, canned, 60 g (¼ cup)	9	Nondairy creamer, 2 Tbsp	9
Lunch		Lunch	
Formula mixture, 6 oz		Formula mixture, 6 oz	
Low-protein bread, ½ slice	6	Vegetable soup, ½ cup	82
Jelly, 1 tsp	0	Grapes, 50 g (10)	9
Carrots, cooked, 40 g (¼ cup)	14	Low-protein crackers, 5	3
Apricots, canned, 25 g (½ cup)	32	Low-protein cookie, 2	8
Snack		Snack	
Apple slice, peeled, ½ cup	3	Rice cakes, 6 g (2 mini)	25
Goldfish crackers, 10	18	Jelly, 1 tsp	0
Formula mixture, 6 oz		Formula mixture, 6 oz	
Dinner		Dinner	
Formula mixture, 6 oz		Formula mixture, 6 oz	
Low-protein pasta, ½ C, cooked	5	Potato, diced, 50 g (5 Tbsp)	39
Tomato sauce, 2 Tbsp	12	Dairy-free margarine, 1 tsp	0
Green beans, cooked, 17 g (2 Tbsp)	9	Zucchini, sauteed, ¼ cup (45 g)	10
<i>Total phenylalanine from food</i>	130 mg	<i>Total phenylalanine from food</i>	210

TABLE 44.5 Comparison of Protein and Energy Content of Foods Used in Low-Protein Diets

Food Item	Energy (kcal)	Protein (g)
Pasta, ½ cup, cooked		
Low-protein	107	0.15
Regular	72	2.4
Bread, 1 slice		
Low-protein	135	0.2
Regular	74	2.4
Cereal, ½ cup, cooked		
Low-protein	45	0.0
Regular	80	1.0
Egg, 1		
Low-protein egg replacer	30	0.0
Regular	67	5.6

**Fig. 44.3** Two young children, both with phenylketonuria, who were identified by a newborn screening program and started on treatment by 7 days of age, demonstrate typical growth and development. (Courtesy Beth Ogata, Seattle, WA.)

TABLE 44.6 Comparison of Menus Appropriate for Children With and Without Phenylketonuria

Meal	Menu for PKU	Phenylalanine (mg)	Regular Menu	Phenylalanine (mg)
Breakfast	Phenylalanine-free formula	0	Milk	450
	Puffed rice cereal	19	Puffed rice cereal	19
	Orange juice	11	Orange juice	11
Lunch	Jelly sandwich with low-protein bread	18	Peanut butter and jelly sandwich with regular bread	625
	Banana	49	Banana	49
	Carrot and celery sticks	12	Carrot and celery sticks	12
	Low-protein chocolate chip cookies	4	Chocolate chip cookies	60
	Juice	0	Juice	0
Snack	Phenylalanine-free formula	0	Milk	450
	Orange	16	Orange	16
	Potato chips (small bag)	44	Potato chips	44
Dinner	Phenylalanine-free formula	0	Milk	450
	Salad	10	Salad	10
	Low-protein spaghetti with tomato sauce	8	Spaghetti with tomato sauce and meatballs	240 600
	Sorbet	10	Ice cream	120
	<i>Estimated intake</i>	<i>201</i>		<i>3156</i>

PKU, Phenylketonuria.

TABLE 44.7 Tasks Expected of Children With Phenylketonuria by Age Level

Age (year)	School Level	Task
2–3	Preschool	Distinguishing between “yes” and “no” foods
3–4	Preschool	Counting: how many?
4–5	Preschool	Measuring: how much?
5–6	Kindergarten	Preparing own formula; using scale
6–7	Grade 1–2	Writing basic notes in food diary
7–8	Grade 2	Making some decisions on after-school snack
8–9	Grade 3	Preparing breakfast
9–10	Grade 4	Packing lunches
10–14	Middle school	Managing food choices with increasing independence
14–18	High school	Independently managing phenylketonuria

perhaps to restrict their social activities. The children, in turn, may react negatively to their parents and to their nutrition therapy. The ability of the family to respond to the stresses of PKU, as reflected by adaptability and cohesion scores, is demonstrated by improved blood-Phe concentrations and the positive coping behaviors of older children with PKU. Continuing nutrition therapy beyond early childhood requires

that children become knowledgeable about and responsible for managing their own food choices. The health care team becomes responsible for working with families and children to provide strategies that enable children and adolescents to participate in social and school activities, interact with peers, and progress through the typical developmental stages with self-confidence and self-esteem.

Children require parental and professional support as they assume responsibility for their food management. Self-management of food choices is a strategy to prevent the child from using dietary noncompliance as a wedge against parental restrictions. Normal intellectual development is a laudable goal of the management of PKU, but to be entirely successful, children with PKU concomitantly need to develop self-assurance and a strong self-image. This can be achieved in part by fostering self-management, problem-solving skills, independence, and a typical lifestyle.

Maternal Phenylketonuria

A pregnant woman with elevated blood-Phe concentrations endangers her fetus because of the active transport of amino acids across the placenta. The fetus is exposed to approximately twice the Phe level contained in normal maternal blood. Babies whose mothers have elevated blood-Phe concentrations have an increased occurrence of cardiac defects, restricted growth, microcephaly, and intellectual disability, as presented in Table 44.8. The fetus appears to be at risk of damage even with minor elevations in maternal blood-Phe levels, and the higher the level, the more severe the effect. The strict control of maternal Phe levels before conception and throughout pregnancy offers the best opportunity for normal fetal development (Koch et al, 2010; Martino et al, 2013).

Nutrition management for pregnant women with hyperphenylalaninemia is complex. The changing physiology of pregnancy and

TABLE 44.8 Frequency of Abnormalities in Children Born to Mothers With Phenylketonuria

Complication (% of Offspring)	MATERNAL PHENYLALANINE LEVELS (mg/dL)				Non-PKU Mother
	20	16–19	11–15	3–10	
Mental retardation	92	73	22	21	5.0
Microcephaly	73	68	35	24	4.8
Congenital heart disease	12	15	6	0	0.8
Low birth weight	40	52	56	13	9.6

(Modified from Lenke RR, Levy HL: Maternal phenylketonuria and hyper-phenylalaninemia: an international survey of the outcome of untreated and treated pregnancies, *N Engl J Med* 303:1202, 1980.)
PKU, Phenylketonuria.

fluctuating nutritional needs are difficult to monitor with the precision required to maintain appropriately low blood-Phe concentrations. Even with meticulous attention to Phe intake, blood concentrations, and the nutrient requirements of pregnancy, a woman cannot be ensured a normal infant (Lee et al, 2005). The risks of abnormal development of the fetus, even with therapeutic dietary management and maintenance of blood-Phe concentrations at 1 to 5 mg/dL (60 to 300 μ mol/L), are an important consideration for young women with PKU considering pregnancy (Waisbren and Azen, 2003).

Nutritional management during pregnancy is challenging, even for women who have consistently followed a low-Phe dietary regimen since infancy. Women who have discontinued Phe dietary treatment find that reinstituting medical food consumption and limiting food choices can be difficult and overwhelming. Inadequate maternal nourishment (i.e., inadequate intakes of total protein, fat, and energy) may contribute to poor fetal development and should be prevented. Adherence to nutrition therapy during pregnancy for even the well-motivated woman requires family and professional support, as well as frequent monitoring of biochemical and nutritional aspects of pregnancy and PKU.

Adults Living With Phenylketonuria

Many adults with PKU have had the benefits of early diagnosis and treatment and are less likely to be affected by neurologic damage. However, among those who have had some degree of intellectual disability, hyperactivity and self-abuse are often major concerns. Not all patients respond to late initiation of treatment with improved behavioral or intellectual function. For the difficult-to-manage older patient, a trial of a low-Phe food pattern is recommended. If successful, continued Phe-restriction therapy may facilitate behavioral management.

Reinstituting a Phe-restricted food pattern is difficult after the eating pattern has been liberalized. However, the current recommendation of most clinics is the effective management of blood-Phe concentration throughout a lifetime. This recommendation is based on reports of declining intellectual capabilities and changes in the brain after a prolonged, significant elevation of Phe concentrations (Camp et al, 2014). The efficacy of continued treatment throughout adulthood has been documented by reports of improved intellectual performance and problem-solving abilities when blood-Phe levels are

kept low. Dietary management of PKU throughout the life span is similar to that of other chronic disorders, and prudent MNT results in a normal quality of life.

Maple Syrup Urine Disease

Maple syrup urine disease (MSUD), or **branched-chain ketoaciduria**, results from a defect in enzymatic activity, specifically the branched-chain α -ketoacid dehydrogenase complex. It is an autosomal-recessive disorder. Infants appear normal at birth, but by 4 or 5 days of age, they demonstrate poor feeding, vomiting, lethargy, and periodic hypotonia. A characteristic sweet, malty odor from the urine and perspiration can be noted toward the end of the first week of life.

Pathophysiology

The decarboxylation defect of MSUD prevents metabolism of the branched-chain amino acids (BCAAs) leucine, isoleucine, and valine (Fig. 44.4). Leucine tends to be more problematic than the others. The precise mechanism for the complete decarboxylase reaction and the resultant neurologic damage is not known. Neither is the reason why leucine metabolism is significantly more abnormal than that of the other two BCAAs (Strauss et al, 2006).

Medical Treatment

Failure to treat this condition leads to acidosis, neurologic deterioration, seizures, and coma, proceeding eventually to death. Management of acute disease often requires peritoneal dialysis and hydration (see Chapter 35).

Depending on the severity of the enzyme defect, early intervention and meticulous biochemical control can provide a more hopeful prognosis for infants and children with MSUD. Reasonable growth and intellectual development in the normal-to-low-normal range have been described. Diagnosis before 7 days of age and long-term metabolic control are critical factors in long-term normalization of intellectual development. Maintenance of plasma leucine concentrations in infants and preschool children should be as close to physiologically normal as possible. Concentrations greater than 10 mg/dL (760 μ mol/L) often are associated with α -ketoacidemia and neurologic symptoms.

Because the liver is the central site of metabolic control for amino acids and other compounds that cause acute degeneration of the brain during illness, therapeutic liver transplantation is sometimes an option in MSUD. Liver transplantation can prevent decompensation and crises associated with elevated leucine levels, but it does not reverse existing neurologic damage (Diaz et al, 2014; Oishi et al, 2016).

Medical Nutrition Therapy

Nutrition therapy requires very careful monitoring of blood concentrations of leucine, isoleucine, and valine as well as growth and general nutritional adequacy. Several formulas specifically designed for the treatment of this disorder are available to provide a reasonable amino acid and vitamin mixture. These generally are supplemented with a small quantity of standard infant formula or cow's milk to provide the BCAAs needed to support growth and development. Some infants and children may require additional supplementation with L-valine or L-isoleucine to maintain biochemical balance.

BCAAs may be introduced gradually into the diet when plasma leucine concentrations are decreased sufficiently (Frazier et al, 2014). Clinical relapse is related most often to the degree of elevation of leucine concentrations, and these relapses often are related to infection. Acute infections represent life-threatening medical emergencies in this group of children. If the plasma leucine concentration increases rapidly during illness, BCAAs should be removed from the diet immediately and intravenous therapy can be started.

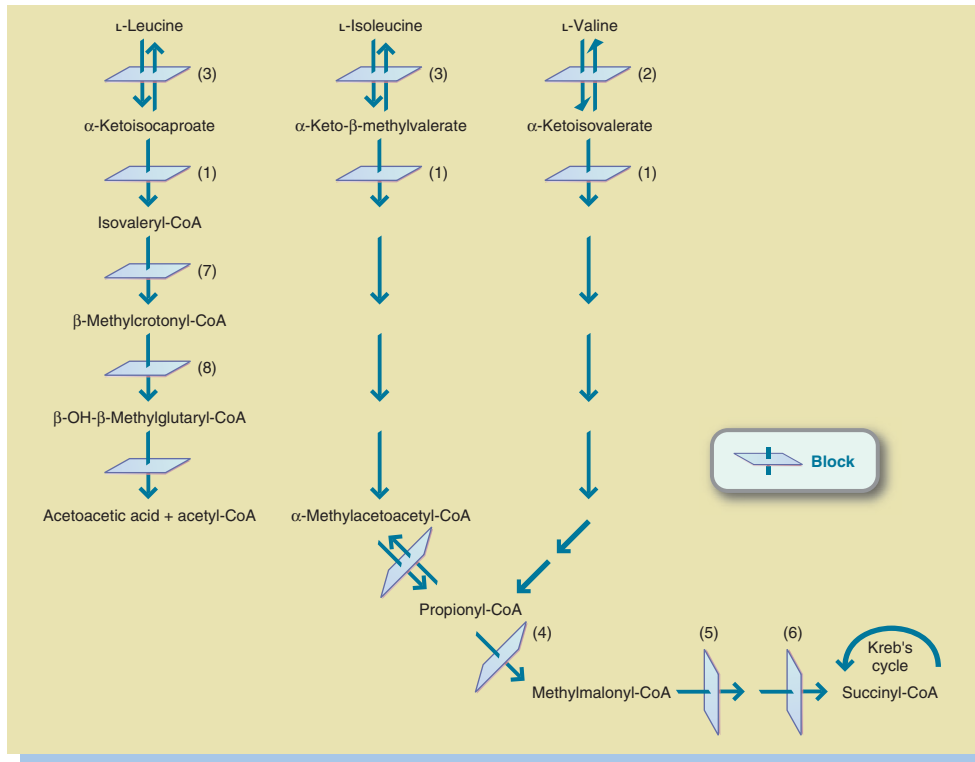


Fig. 44.4 Organic acidemias and maple syrup urine disease (*MSUD*). (1) Branched-chain keto-acid decarboxylase (*MSUD*); (2) Valine aminotransferase; (3) Leucine-isoleucine aminotransferase; (4) Propionyl-CoA carboxylase (propionic acidemia); (5) Methylmalonyl-CoA racemase (methylmalonic aciduria); (6) Methylmalonyl-CoA mutase (methylmalonic aciduria); (7) Isovaleryl-CoA dehydrogenase (isovaleric acidemia); (8) β-methylcrotonyl-CoA carboxylase (biotin-responsive multiple carboxylase deficiency). CoA, Coenzyme A.

DISORDERS OF ORGANIC ACID METABOLISM

Organic acid disorders are a group of disorders characterized by the accumulation in the blood of nonamino acid organic acids. Usually, most of the organic acids are excreted efficiently in the urine. Diagnosis is based on the excretion of compounds not normally present or the presence of abnormally high amounts of other compounds in the urine. The clinical course can vary but is generally marked by vomiting, lethargy, hypotonia, dehydration, seizures, and coma. Survivors often have permanent neurologic damage.

Pathophysiology

Propionic acidemia is a defect of propionyl-coenzyme A (CoA) carboxylase in the pathway of propionyl-CoA to methylmalonyl-CoA, as illustrated in Fig. 44.4. Metabolic acidosis with a marked anion gap and hyperammonemia is characteristic, and long-chain ketonuria also may be present.

At least five separate enzyme deficiencies have been identified that result in **methylmalonic acidemia** or aciduria. The defect of methylmalonyl-CoA mutase apoenzyme is the most frequently identified. In methylmalonic acidemia, the clinical features are similar to those of propionic acidemia. Acidosis is common, and diagnosis is confirmed by the presence of large amounts of methylmalonic acid in blood and urine. Other findings include hypoglycemia, ketonuria, and elevation of plasma ammonia and lactate levels.

Ketone utilization disorders (mitochondrial 2-methylacetoacetyl-CoA thiolase deficiency or similar enzyme defect) are disorders of isoleucine and ketone body metabolism. Affected individuals are usually

older infants or toddlers who present with ketoacidosis, vomiting, and lethargy with secondary dehydration and sometimes coma. This event often is preceded by febrile illness or fasting.

Medical Treatment

Some patients with propionic acidemia may respond to pharmacologic doses of biotin. Long-term outcome in propionic acidemia is variable; hypotonia and cognitive delay may result even in children who are diagnosed early and who receive rigorous treatment. Liver damage and cardiomyopathy are possible sequelae. Liver transplantation may limit intellectual disability and cardiac changes (Sutton et al, 2012).

Individuals with methylmalonic acidemia may respond to pharmacologic doses of vitamin B₁₂. Responsiveness should be determined as part of the diagnostic process (Manoli et al, 2016). Progressive renal insufficiency is often a long-term outcome. Developmental delay is often caused by early and/or prolonged hyperammonemia.

For ketone utilization disorders, the treatment is dietary protein restriction (usually 1.5 g/kg of body weight per day); supplementation with **L-carnitine**, a carrier of fatty acids across the mitochondrial membranes; avoidance of fasting by providing small, frequent meals that consist primarily of complex carbohydrates; and the use of Bicitra (sodium citrate-citric acid) to treat ketoacidosis.

Medical Nutrition Therapy

The goals of managing acute episodes of propionic acidemia and methylmalonic acidemia are to achieve and maintain normal nutrient intake and biochemical balance. Maintenance of energy and fluid intake is important to prevent tissue catabolism and dehydration. Intravenous

fluids correct electrolyte imbalances, and abnormal metabolites are removed through urinary excretion, promoted by a high-fluid intake. Relapses of metabolic acidosis may result from excessive protein intake, infection, constipation, or unidentified factors. Treatment for these episodes must be rapid because coma and death can occur quickly. Parents become skilled at identifying early signs of illness ([Southeast Regional Genetics Network \[SRGN\] and GMDI, 2017](#)).

Restricted-protein intake is an essential component of the treatment of organic acid disorders. A daily protein intake of 1 to 1.5 g/kg of body weight is often an effective treatment modality for infants who have a mild form of the disorder. This can be supplied by diluting standard infant formula to decrease the protein content and adding a protein-free formula to meet other nutrient needs. Specialized formulas that limit threonine and isoleucine and omit methionine and valine are used, as clinically indicated, to support an adequate protein intake and growth.

Requirements for the limited amino acids may vary widely. Growth rate, state of health, residual enzyme activity, and overall protein and energy intakes must be monitored carefully and correlated with plasma amino acid levels. Adequate hydration is critical to maintain metabolic equilibrium. Food refusal and lack of appetite may complicate nutrition therapy, compromising medical management.

DISORDERS OF UREA CYCLE METABOLISM

All **urea cycle disorders (UCDs)** result in an accumulation of ammonia in the blood. The clinical signs of elevated ammonia are vomiting and lethargy, which may progress to seizures, coma, and ultimately death. In infants, the adverse effects of elevated ammonia levels are rapid and devastating. In older children, symptoms of elevated ammonia may be preceded by hyperactivity and irritability. Neurologic damage may result from frequent and severe episodes of hyperammonemia. The severity and variation of the clinical courses of some urea cycle defects may be related to the degree of residual enzyme activity. The common urea cycle defects are discussed in a progression that proceeds around the urea cycle, as shown in [Fig. 44.5](#).

Pathophysiology

Ornithine transcarbamylase (OTC) deficiency is an X-linked disorder marked by blockage in the conversion of ornithine and carbamyl phosphate to citrulline. OTC deficiency is identified by hyperammonemia and increased urinary orotic acid, with normal levels of citrulline, argininosuccinic acid, and arginine. Severe OTC deficiency is usually lethal in males. Heterozygous females with various degrees of enzyme activity may not demonstrate symptoms until they are induced by stress, as from an infection, or a significant increase in protein intake.

Citrullinemia is the result of a deficiency of argininosuccinic acid synthetase in the metabolism of citrulline to argininosuccinic acid. Citrullinemia is identified by markedly elevated citrulline levels in the urine and blood. Symptoms may be present in the neonatal period, or they may develop gradually in early infancy. Poor feeding and recurrent vomiting occur which, without immediate treatment, progress to seizures, neurologic abnormalities, and coma.

Argininosuccinic aciduria (ASA) results from a deficiency of argininosuccinate lyase, which is involved in the metabolism of argininosuccinic acid to arginine. ASA is identified by the presence of argininosuccinic acid in urine and blood. L-arginine must be supplemented to provide an alternative pathway for waste nitrogen excretion.

Carbamyl-phosphate synthetase (CPS) deficiency is the result of a deficient activity of CPS. The onset is usually in the early neonatal period, with vomiting, irritability, marked hyperammonemia, respiratory distress, altered muscle tone, lethargy, and often coma. Specific laboratory findings usually include low-plasma levels of citrulline and arginine and normal orotic acid levels in urine.

Medical Treatment

Acute episodes of illness are managed by discontinuing protein intake and administering intravenous fluids and glucose to correct dehydration and provide energy. If hyperammonemia is severe, peritoneal dialysis, hemodialysis, or exchange transfusion may be required. Intravenous sodium benzoate or other alternative pathway compounds have been beneficial in reducing the hyperammonemia.

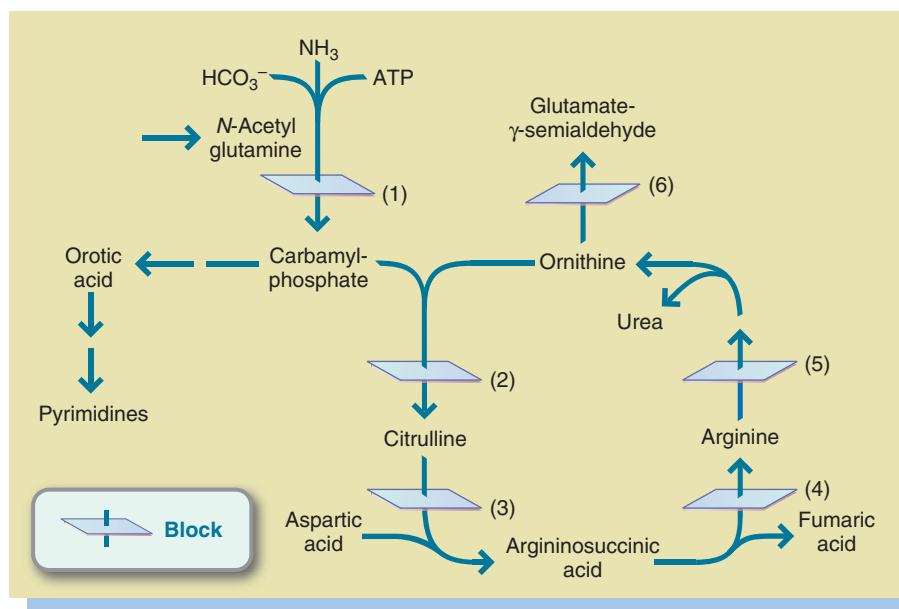


Fig. 44.5 Urea cycle disorders (UCD). (1) Carbamyl-phosphate synthetase deficiency; (2) Ornithine carbamyl transferase deficiency; (3) Argininosuccinic acid synthetase (citrullinemia); (4) Argininosuccinic acid lyase (argininosuccinic aciduria); (5) Arginase deficiency (argininemia); (6) Adenosine triphosphatase.

Neurologic outcome and intellectual development in individuals with UCDs vary, with a range from normal IQ and motor function to severe intellectual disability and cerebral palsy. Although information on long-term follow-up is limited, the use of alternative pathways for waste nitrogen excretion and a protein-restricted food pattern may improve the outcome.

Medical Nutrition Therapy

Nutritional management of patients who have UCDs is a challenging task. The aim of therapy for the UCDs is to prevent or decrease hyperammonemia and the detrimental neurologic consequences associated with it. Treatment is similar for all of the disorders. For mildly affected infants, a standard infant formula can be diluted to provide protein at 1 to 1.5 g/kg body weight per day. The energy, vitamin, and mineral concentrations can be brought up to recommended intake levels with the addition of a protein-free formula that contains these nutrients. However, for most individuals, specialized formulas are needed to adjust protein composition in an effort to limit ammonia production.

The amount of protein tolerated is affected by variables such as the specific enzyme defect, age-related growth rate, health status, level of physical activity, amount of free amino acids administered, energy needs, residual enzyme function, and the use of nitrogen-scavenging medications. Recommendations must take family lifestyle and the individual's eating behaviors into consideration. Long-term therapy consists of restricting dietary protein to 1 to 2 g/kg/day, depending on individual tolerance. For most infants and children with these disorders, L-arginine supplements are required to prevent arginine deficiency and assist in waste nitrogen excretion. L-arginine is supplemented based on individual needs, except in the case of arginase deficiency (Brusilow and Horwich, 2010). Phenylbutyrate or other compounds that enhance alternative metabolic pathways usually are required to normalize ammonia levels.

Protein-Restricted Diets

Infants and children with urea cycle defects or organic acidemias generally require restricted-protein intakes and specialized formulas. The amount of protein prescribed is based on the individual's tolerance or residual enzyme activity, age, and projected growth rate. The highest protein level tolerated should be given to ensure adequate growth and a margin of nutritional safety. The steps for effective planning of a low-protein food pattern are shown in Box 44.1.

BOX 44.1 Steps in Designing a Low-Protein Eating Plan

1. Determine the protein tolerance of the individual based on (1) diagnosis, (2) age, and (3) growth. Consider the metabolic stability and total protein intake required for the infant's or child's weight.
2. Calculate the protein and energy needs of the individual based on age, activity, and weight.
3. Provide at least 70% of total protein as high-biologic value protein from formula for infants, and from milk or dairy foods for older children. Use a specialized formula if the infant or child cannot tolerate the entire protein intake from intact protein.
4. Provide energy and nutrient sources to meet basic needs.
5. Add water to meet fluid requirements and maintain appropriate concentration of formula mixture.
6. For the older infant and child, provide foods to meet food variety, texture, and energy needs.
7. Provide adequate intake of calcium, iron, zinc, and all other vitamins and minerals for age.

In general, low-protein or restricted-protein food patterns can be formulated from readily available, lower-protein infant, toddler, and table foods. Special low-protein foods (see Table 44.5) can be used to provide energy, texture, and variety in the food pattern without appreciably increasing the protein load. The prescribed protein level can be met by adding a protein-free or specialized formula product to standard infant formula. Supplementing carbohydrates and fat makes up the resultant energy deficit.

Specialty formulas are available when needed. The appropriate choice depends on the level of protein restriction, age, and condition of the child. The usual recommendations for energy density and vitamin and mineral composition are generally appropriate to support growth for the infant or child. The osmolality of the formula and the individual's ability to tolerate hyperosmolar solutions must be considered.

DISORDERS OF CARBOHYDRATE METABOLISM

Disorders of carbohydrate metabolism vary in presentation, clinical course, and outcome. Galactosemia may present in the early newborn period as life-threatening seizures and sepsis. Hereditary fructose intolerance may present during midinfancy when solids that contain offending ingredients are introduced. Glycogen storage diseases may present at the time when feedings are spaced and subsequent hypoglycemia appears. All of these disorders require early and aggressive nutritional therapy.

Hereditary Fructose Intolerance

Hereditary fructose intolerance (HFI) results from a deficiency of the liver enzyme aldolase B. The disorder typically presents in infancy and toddlerhood, when dietary sources of sucrose and fructose are introduced. Initial symptoms include nausea, bloating, and vomiting. Untreated, liver and kidney damage can occur, along with growth restriction.

Medical Nutrition Therapy

Current management for HFI is the restriction of dietary sources of fructose (including disaccharides composed of fructose) and some sugar alcohols, such as sorbitol. HFI (incidence of about 1 in 20,000 to 30,000 in Europe) is different than the more common fructose malabsorption, in which individuals experience gastrointestinal symptoms after ingesting large amounts of fructose (Baker et al, 2015) (see Chapter 28).

Galactosemia

Galactosemia, an elevated level of plasma galactose-1-phosphate combined with galactosuria, is found in two autosomal-recessive metabolic disorders: **galactokinase deficiency** and **galactose-1-phosphate uridylyltransferase (GALT) deficiency**, which is also called "classic galactosemia." Illness generally occurs within the first 2 weeks of life. Symptoms are vomiting, diarrhea, lethargy, failure to thrive, jaundice, hepatomegaly, and cataracts. Infants with galactosemia may be hypoglycemic and susceptible to infection from gram-negative organisms. If the condition is not treated, death frequently ensues secondarily to septicemia.

Pathophysiology

Galactosemia results from a disturbance in the conversion of galactose to glucose because of the absence or inactivity of one of the enzymes shown in Fig. 44.6. The enzyme deficiency causes an accumulation of galactose, or galactose and galactose-1-phosphate, in body tissues. In addition, expanded newborn screening programs have identified many newborns with Duarte galactosemia. These infants have one allele for

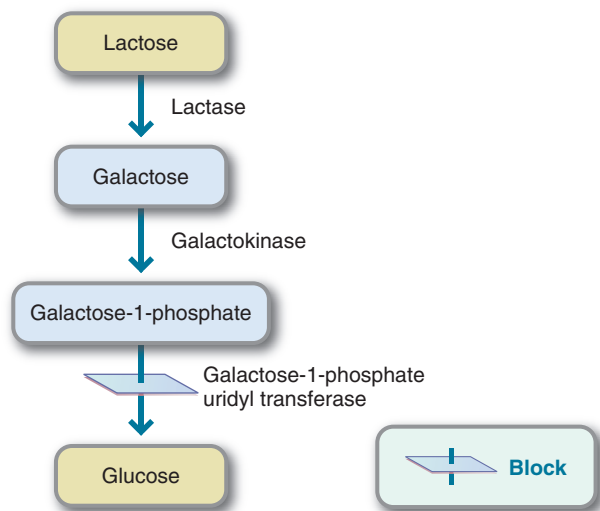


Fig. 44.6 Schematic diagram of the metabolism of galactose in galactosemia.

galactosemia and one for Duarte galactosemia and are often said to have “D/G galactosemia.” The Duarte allele produces approximately 5% to 20% of the GALT enzyme. Little is known about the natural history of D/G galactosemia; apparently, infants and children develop normally without medical complications.

Medical Treatment

When diagnosis and therapy are delayed, intellectual disability can result (Berry, 2017). With early diagnosis and treatment, physical and motor development should proceed normally. However, intellectual achievement may be depressed. Patients often have IQs of 85 to 100, visual-perceptual and speech difficulties, and problems with executive function (Doyle et al, 2010; Kaufman et al, 1995). Ovarian failure affects approximately 75% to 95% of women with galactosemia (Forges et al, 2006).

Medical Nutrition Therapy

Galactosemia is treated with lifelong galactose restriction (van Calcar et al, 2014). Although galactose is required for the production of galactolipids and cerebroside, it can be produced by an alternative pathway if galactose is omitted from the diet. Galactose restriction mandates strict avoidance of all milk and milk products and lactose-containing foods because lactose is hydrolyzed into galactose and glucose. Effective galactose restriction requires careful reading of food product labels. Milk is added to many products, and lactose often appears in the coating of the tablet form of medications. Infants with galactosemia are fed soy-based formula. Some fruits and vegetables contain significant amounts of galactose. Table 44.9 presents a low-galactose food pattern.

Medical opinions differ about the intensity and duration of treatment for Duarte galactosemia (Berry, 2017). Many centers eliminate galactose from the diets of these children for the first year of life; other centers do not.

Glycogen Storage Diseases

Glycogen storage diseases (GSDs) reflect an inability to metabolize glycogen to glucose. There are a number of possible enzyme defects along the pathway. The most common of the GSDs are types I and III. Their symptoms are poor physical growth, hypoglycemia,

TABLE 44.9 Food Lists for Low-Galactose Food Pattern

Allowable Foods	Galactose-Containing Foods to Be Avoided ^a
Milk and Milk Substitutes Similac Soy Isomil Enfamil ProSobee Gerber Good Start Soy	All forms of animal milk Imitation or filled milk Cream, butter, some margarine Cottage cheese, cream cheese Hard cheeses Yogurt Ice cream, ice milk, sherbet Breastmilk
Fruits All fresh, frozen, canned, or dried fruits except those processed with unsafe ingredients ^b	
Vegetables All fresh, frozen, canned, or dried vegetables except those processed with unsafe ingredients, ^b seasoned with butter or margarine, breaded, or creamed	
Meat, Poultry, Fish, Eggs, Nuts Plain beef, lamb, veal, pork, ham, fish, turkey, chicken, game, fowl, Kosher frankfurters, eggs, nut butters, nuts	
Breads and Cereals Cooked and dry cereals, bread or crackers without milk or unsafe ingredients, ^b macaroni, spaghetti, noodles, rice, tortillas	
Fats All vegetable oils; all shortening, lard, margarine, and salad dressings except those made with unsafe ingredients ^b ; mayonnaise; olives	

^aLactose is often used as a pharmaceutical bulking agent, filler, or excipient; thus tablets, tinctures, and vitamin and mineral mixtures should be evaluated carefully for galactose content. The *Physician's Desk Reference* now lists active and inactive ingredients in medications, as well as manufacturers' telephone numbers.

^bUnsafe ingredients include milk, buttermilk, cream, lactose, galactose, casein, caseinate, whey, dry milk solids, or curds. Labels should be checked regularly and carefully because formulations of products change often (see Chapter 26).

hepatomegaly, and abnormal biochemical parameters, especially for cholesterol and triglycerides. Advances in the treatment of GSDs may improve the quality of life for affected children (Weinstein et al, 2018).

Pathophysiology

GSD type Ia is a defect in the enzyme glucose-1,6-phosphatase, impairing the formation of new glucose (**gluconeogenesis**) and the

breakdown of glycogen from storage (**glycogenolysis**). The affected person is unable to metabolize glycogen stored in the liver. Severe hypoglycemia can result and cause irreparable neurologic damage.

Amylo-1,6-glucosidase deficiency (GSD III or debrancher enzyme deficiency) prevents glycogen breakdown beyond branch points. This disorder is similar to GSD Ia, in that glycogenolysis is inefficient but gluconeogenesis is amplified to help maintain glucose production. The symptoms of GSD III are usually less severe and range from hepatomegaly to severe hypoglycemia.

Medical Treatment

The outcome of treatment has been good. The hazard of severe hypoglycemic episodes is diminished, physical growth is improved, and liver size is decreased. The risk of progressive renal dysfunction is not entirely eliminated by current treatment, but liver transplantation for some types of GSD (e.g., type Ib) is sometimes an option. Treatment guidelines include various kinds of carbohydrates at various doses during the day and night (Kishnani et al, 2014). Individual tolerance, body weight, state of health, ambient temperature, and physical activity are important considerations when designing the specific pattern of carbohydrate administration. The goal for all of the protocols remains the same: normalization of blood glucose levels.

Medical Nutrition Therapy

The rationale for intervention is to maintain plasma glucose in a normal range and prevent hypoglycemia by providing a constant supply of exogenous glucose. Administration of raw cornstarch (e.g., a slurry of cornstarch mixed with cold water) at regular intervals and a high-complex carbohydrate, low-fat dietary pattern is advocated to prevent hypoglycemia. Some infants and children do very well with oral cornstarch administration, whereas others require glucose polymers administered via continuous-drip gastric feedings to prevent hypoglycemic episodes during the night. The dose of cornstarch should be individualized; doses of 1.6 to 2.5 g/kg at 4- to 6-hour intervals are generally effective for young children with GSD I (Kishnani et al, 2014). A modified (extended-release) cornstarch product (Glycosade) is also used to manage GSD, typically used at night with older children and adults (Weinstein et al, 2018). Iron supplementation is required to maintain adequate hematologic status because cornstarch interferes with iron absorption.

DISORDERS OF FATTY ACID OXIDATION

Laboratory advancements have enabled the identification of **fatty acid oxidation disorders** such as **medium-chain acyl-CoA dehydrogenase (MCAD) deficiency** and **long-chain 3-hydroxyacyl-CoA dehydrogenase (LCHAD) deficiency** (Fig. 44.7). Children who are not identified by newborn screening usually present during periods of fasting or illness with symptoms of variable severity, including failure to thrive, episodic vomiting, and hypotonia.

Pathophysiology

Children with MCAD deficiency who present clinically typically have hypoglycemia without urine ketones, lethargy, seizures, and coma. Children with LCHAD deficiency become hypoglycemic and demonstrate abnormal liver function, reduced or absent ketones in the urine, and often secondary carnitine deficiency. They also may have hepatomegaly and acute liver disease. Hypoglycemia can progress quickly and be fatal (Matern and Rinaldo, 2015). Both disorders are included in the newborn screening panels in all US states.

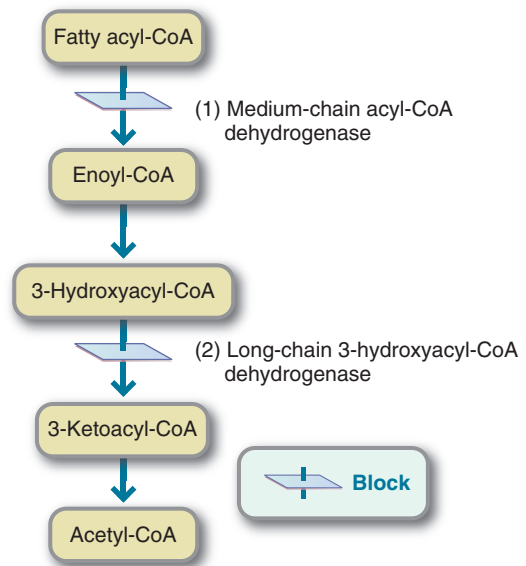


Fig. 44.7 Mitochondrial fatty acid oxidation disorders: (1) Medium-chain acyl-Coenzyme A dehydrogenase deficiency, the most common fatty acid oxidation disorder; (2) Long-chain 3-hydroxyacyl-CoA dehydrogenase deficiency.

Medical Nutrition Therapy

The concept underlying effective treatment for fatty acid oxidation disorders is straightforward: avoidance of fasting. This is accomplished by the regularly spaced intake of foods that provide an adequate energy intake and are high in carbohydrates. A low-fat diet is advocated because fats are not effectively metabolized. Consumption of not more than 30% of energy as fat has been recommended; some individuals require more restriction. Supplementation with L-carnitine, a substance that functions as a carrier of fatty acids across the mitochondrial membranes, is recommended by some clinics.

Children often do very well with three meals and three snacks offered at regular intervals. Most children may require additional carbohydrates before bed based on individual ability to maintain blood glucose levels throughout the night (Matern and Rinaldo, 2015). Depending on the disorder, supplementation with specific fatty acids (e.g., medium-chain fats for disorders that involve blocks of long-chain metabolism) may be indicated.

ROLE OF THE NUTRITIONIST IN GENETIC METABOLIC DISORDERS

The role of the pediatric nutrition specialist in the treatment of metabolic disorders is a complex one that requires expertise in MNT for the specific disorder. Preparation and competency require access to detailed information about the disorders and treatment modalities. A family-centered approach, knowledge of feeding-skill development, and understanding of behavior modification techniques, as well as the support and counsel of a team of health care providers involved in the care of the patient, are also required. Nutrition intervention is often a lifelong consideration. Specific objectives of nutrition care are shown in Box 44.2.

BOX 44.2 Intervention Objectives for the Nutritionist Involved in the Treatment of Genetic Metabolic Disorders

In the clinic the registered dietitian nutritionist (RDN) has a major role in ongoing therapy and planning for each child. These responsibilities include gathering of objective food intake data from the family, assessing the adequacy of the child's intake, and working with the family to incorporate appropriate ways to monitor the restricted-food intake pattern. The child with a metabolic disorder often presents with a wide range of concerns, which may include unstable biochemical markers, failure to gain weight, excessive weight gain, difficulty adhering to the diet, and behaviors that cause an adverse feeding situation. Thus, managing a child with a metabolic disorder requires input from the entire health care team. The nutritionist uses skills and knowledge of foods as sources of nutrients, parent-child relationships, growth, development, and interviewing to obtain the necessary information for assessing and planning for the child with a genetic metabolic disorder.

- I. The RDN functions as an effective interdisciplinary team member by doing the following:
 - A. Becoming familiar with the background and current status of the child by reviewing the medical record
 - B. Recognizing and accepting the responsibility as the nutritionist by doing the following:
 1. Identifying appropriate intake of nutrients for growth, activity, and biochemical balance
 2. Identifying developmental stages of feeding behavior
 3. Understanding the concept of food as a support of developmental progress
 4. Identifying behavior as it affects nutrient intake
 - C. Understanding, respecting, and using the expertise of the team disciplines in providing care for the child
- II. The RDN provides adequate and supportive patient services by doing the following:
 - A. Establishing a positive, cooperative working relationship with the parent, child, and other family members
 - B. Interviewing the family about dietary intake and the feeding situation in a nonjudgmental manner
 - C. Assessing the parent-child relationship as it relates to dietary management and control of the disorder
 - D. Developing a plan for appropriate dietary management based on growth, biochemical levels, nutrient needs, developmental progress, and nutrition diagnosis, such as:
 - Excessive protein intake
 - Altered nutrition-related laboratory values
 - Intake of inappropriate amino acids
 - Inadequate vitamin intake
 - Inadequate mineral intake
 - Food-medication interaction
 - Food- and nutrition-related knowledge deficit
 - Limited adherence to nutrition-related recommendations
 - E. Developing a plan that includes appropriate foods and recognizes the family's skills in food preparation, as well as family routines
 - F. Working with the family to establish a method to deal effectively with negative feeding behaviors, if necessary
 - G. Contacting the family after receiving laboratory results and calculating food records to make necessary and appropriate changes in diet prescription
- III. Supporting families in their efforts at effective dietary and behavior management
- IV. The RDN develops a professional knowledge base by doing the following:
 - A. Being familiar with the current literature on the treatment of metabolic disorders
 - B. Understanding the genetic basis of metabolic disorders
- V. The RDN works with team members to understand the long-term patient care and create a written care plan for the patient.

CLINICAL CASE STUDY

Adelaide was born weighing 6 lb, 7 oz and her 1-day newborn screening test result for phenylalanine (Phe) was 3.7 mg/dL (222 μ mol/L). She was breastfed with no supplemental formula. A repeat sample was requested to further document the Phe concentration in her blood. The result from this sample, collected on day 4 of life, was 6.2 mg/dL (372 μ mol/L). To confirm the diagnosis for this child, which was considered to be "presumptive positive," a quantitative sample was obtained, and Phe and tyrosine levels were both measured. On day 9 of life, the serum Phe concentration was 16.6 mg/dL (1328 μ mol/L) and the tyrosine level was 1.1 mg/dL (60.5 μ mol/L); the Phe-to-tyrosine ratio was 22.0:1. Blood and urine were collected for bipterin screening; results were later found to be normal.

To provide adequate protein and energy intake and at the same time decrease the serum Phe concentration, a Phe-free formula was introduced at standard dilution without a Phe supplement. Within 24 hours, Adelaide's serum Phe concentration had decreased to 8.3 mg/dL (498 μ mol/L) while she was being provided an intake of 12 to 14 oz of the Phe-free formula. Breastfeedings were reintroduced; she was offered 3 breastfeedings and Phe-free formula ad lib. Within 48 hours, the level was 6.6 mg/dL (396 μ mol/L), with an intake of 4 oz of the formula.

Adelaide's Phe concentrations were measured every 4 days, and the levels were 3.6 mg/dL (216 μ mol/L) and 2.2 mg/dL (132 μ mol/L). In subsequent weeks growth and serum Phe concentrations continued to be monitored carefully, and energy and Phe intakes were adjusted as necessary to maintain blood-Phe

concentrations between 2 and 6 mg/dL (120 to 360 μ mol/L) and to maintain growth in appropriate channels.

By age 2 months, Adelaide's intake had increased to approximately 12 oz formula along with 3 breastfeedings daily (alternating breastmilk and formula feedings), and her feeding pattern was fairly consistent. Blood-Phe was measured at 8.7 mg/dL (522 μ mol/L), which is outside the desired range. Her Phe-free formula was increased to 14 oz/day to effectively decrease the amount of Phe provided by breastmilk.

Nutrition Diagnostic Statement

- Altered nutrition-related laboratory values (phenylalanine) related to phenylketonuria and dietary Phe intake, as evidenced by blood-Phe outside the desired range.

Nutrition Care Questions

1. What is the expected energy requirement for Adelaide?
2. What are the baseline expectations for intake for a 6-lb, 7-oz neonate? If breastmilk is expected to provide about half of this amount, how much Phe-free formula would you use to provide protein and energy intakes at recommended levels?
3. What are the growth expectations for Adelaide?
4. What steps would you take if Adelaide's plasma Phe concentration exceeded 6 mg/dL (360 μ mol/L) on subsequent measurements?

USEFUL WEBSITES

American College of Medical Genetics (ACMG)
 Gene Reviews from the National Library of Medicine
 Genetic Metabolic Dietitians International (GMDI)
 Genetics Home Reference from the National Institutes of Health
 MedlinePlus: Metabolic Disorders
 National Newborn Screening and Genetics Resource Center
 National PKU Alliance
 National PKU News

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Medical Nutrition Therapy for Intellectual and Developmental Disabilities

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KEY TERMS

alcohol-related birth defects (ARBDs)
alcohol-related neurodevelopmental disorders (ARNDs)
Arnold-Chiari malformation of the brain
Asperger syndrome
attention-deficit/hyperactivity disorder (ADHD)
autism spectrum disorders (ASDs)
cerebral palsy (CP)

cleft lip and/or cleft palate (CL/CP)
developmental disability
Down syndrome (DS)
fetal alcohol spectrum disorders (FASDs)
hypotonia
individualized education plan (IEP)
individualized family plan
intellectual disability
midfacial hypoplasia

mosaicism
myelomeningocele (MM)
nondisjunction
oral-motor problems
pervasive developmental disorder (PDD)
Prader-Willi syndrome (PWS)
spina bifida
translocation

Individuals with developmental disabilities were generally housed in institutions for the first half of the 20th century. Little attention was paid to their education or medical or nutritional care. In 1963, the Developmental Disabilities Assistance and Bill of Rights Act was passed. Through this Act, federal funds supported the development and operation of state councils, protection and advocacy systems, university centers, and projects of national significance. This act provided the structure to assist people with developmental disabilities to pursue meaningful and productive lives. The institutions that housed these individuals gradually were closed or reduced in size. By 1975, these individuals were cared for at home, in schools, or in small residential facilities. In 1975, Public Law (P.L.) 94–142 was passed, opening public schools to children with developmental disabilities. In 1985, P.L. 99–487 (102–119 in 1992), the Early Intervention Act, was passed, bringing services to children from birth to school age. In 2004, the Individuals with Disabilities Education Act (IDEA) reauthorized this legislation.

A **developmental disability** is defined as a severe chronic disability that is attributable to an intellectual or physical impairment or combination of intellectual and physical impairments. It is manifested before the person reaches age 22; is likely to continue indefinitely; results in substantial functional limitations in three or more areas of major life activity (self-care, receptive and expressive language, learning, mobility, self-direction, capacity for independent living, and economic self-sufficiency); and reflects the person's need for a combination of generic or specialized interdisciplinary care, treatments, or other services that are lifelong or of extended duration and individually planned and coordinated. Developmental disabilities affect individuals of all ages and are not a disease state. Intellectual disability replaced the term *mental retardation* in the 11th edition of the *Definition Manual of the American Association on Intellectual and Developmental Disabilities (AAIDD, 2013)*. **Intellectual disability** is the most common developmental disability, characterized by significantly below-average intellectual functioning along with related limitations in areas such as communication, self-care, functional academics, home living, self-direction, health and safety, leisure, or work and social skills. An estimated 1% to 3% of the population have this diagnosis.

Developmental disabilities have been traced to many causes: chromosomal aberrations, congenital anomalies, specific syndromes, neuromuscular dysfunction, neurologic disorders, prematurity, cerebral palsy (CP), untreated inborn errors of metabolism, toxins in the environment, and nutrient deficiencies. The Centers for Disease Control and Prevention (CDC) reported in 2018 that one in seven, or 15%, of American children have a developmental disability.

MEDICAL NUTRITION THERAPY

Medical nutrition therapy (MNT) services vary depending on the individual, and much has been learned about the role of nutrition in the prevention of disabilities and interventions. The role of the registered dietitian nutritionist (RDN) is essential. Because there is an abundance of information that parents and caretakers use from support groups and websites that are untested scientifically, RDNs often are providing evidence-based counseling to provide support and counteract misinformation.

Numerous nutrition problems have been identified in the individual with developmental disabilities. Growth restriction, obesity, failure to thrive, feeding problems, metabolic disorders, gastrointestinal (GI) disorders, drug-nutrient interactions, constipation, and cardiac and renal problems may be present. Other health problems exist, depending on the disorder. [Table 45.1](#) lists the most common developmental disabilities and their associated nutrition problems.

The Academy of Nutrition and Dietetics confirms that nutrition services provided by RDNs are essential components of comprehensive care for infants and children with special health care needs and adults with intellectual and developmental disabilities (IDDs). Nutrition services should be provided throughout the life cycle in a manner that is interdisciplinary, family centered, community based, and culturally competent. Poor nutrition-related health habits, limited access to services, and long-term use of multiple medications are considered health risk factors in this population. Timely and cost-effective nutrition interventions promote health maintenance, reduce risk, and reduce the cost of comorbidities and complications ([Ptomey and Wittenbrook, 2015](#)).

TABLE 45.1 Common Nutrition-Related Issues in Individuals With Intellectual and Developmental Disabilities and Children and Youth With Special Health Care Needs With Suggested Nutrition Diagnosis Terms

Syndrome or Developmental Disability	Nutrition Diagnosis	Etiology and/or Signs and Symptoms
Autism spectrum disorders (ASDs) ASD is a group of developmental disabilities that are characterized by delayed speech and language development, ritualistic or repetitive behaviors, and impairments in social interactions. Individuals may also have delays in feeding skills and the ability to self-feed.	Inadequate energy intake	Limited food choices Medication side effects affecting appetite
	Limited food acceptance	Sensory processing issues Avoidance of foods/food groups
	Inadequate intake of calcium/vitamin D/iron/other nutrient(s)	Limited food choices; avoidance of foods/food groups Following a special diet such as a gluten-free, casein-free diet, may place child at risk for nutrient deficiencies, including calcium and fiber if food acceptance is already low.
	Underweight	Inadequate energy intake BMI <5th percentile for children aged 2–20 years Medication side effects affecting appetite Limited food choices secondary to behaviors
	Overweight/obesity	BMI >85th percentile for children aged 2–20 years Use of food for behavioral interventions Estimated excessive energy intake Medication side effects affecting appetite Low physical activity level Limited food choices that are excessive in calories
Cerebral palsy (CP) A nonprogressive disorder of muscle control or coordination that affects different parts of the body. This results from an injury to the brain during fetal, perinatal, or early childhood development. The severity is variable. Intellectual disability is often present.	Increased energy expenditure	Unintended weight loss
	Excessive energy intake	Increased body adiposity Reduced energy expenditure secondary to treatment with medication to reduce muscle tone Hypotonia Weight gain greater than expected Changes in mobility status Enteral nutrition provides more calories than measured/estimated energy expenditure
	Inadequate oral intake	Unintended weight loss Inability to self-feed due to lack of coordination Poor dentition, presence of cavities, and/or abscesses Gastroesophageal reflux disease Oral-motor dysfunction/dysphagia Medications affecting appetite Malnutrition and failure to thrive
	Inadequate fluid and fiber intake	Constipation Inability to independently consume fluid and food Dysphagia Altered thirst sensation, inability to communicate thirst
	Swallowing difficulty	Abnormal swallow study showing aspiration and/or oral/pharyngeal dysphagia Coughing, choking, prolonged chewing, pocketing of food, regurgitating, and facial expression changes during eating Reduced food intake Unintended weight loss Prolonged feeding times, lack of interest in food, food avoidance, mealtime resistance
	Altered GI function	Constipation Gastroesophageal reflux disorder, delayed gastric emptying Medication side effects
	Food-medication interactions	Constipation related to antispasticity medications Increased risk of osteopenia/osteoporosis related to antiseizure and antigastroesophageal reflux disease medications Risk of B ₁₂ deficiency related to antigastroesophageal reflux disease medications

TABLE 45.1 Common Nutrition-Related Issues in Individuals With Intellectual and Developmental Disabilities and Children and Youth With Special Health Care Needs With Suggested Nutrition Diagnosis Terms—Cont'd

Syndrome or Developmental Disability	Nutrition Diagnosis	Etiology and/or Signs and Symptoms
	Underweight	BMI <5th percentile/age for children aged 2–20 years Reduced muscle mass, muscle wasting Inadequate energy intake to promote weight gain compared with estimated or measured needs Hypertonia, dystonia
	Overweight	BMI >85th percentile/age for children aged 2–20 years Excessive energy intake Low physical activity level Hypotonia
Drug exposed and fetal alcohol spectrum disorders Physical, behavioral, mental, or learning disabilities that occur when a fetus is exposed to drugs and/or alcohol during development. Growth may also be affected.	Food-medication interaction	Side effects of medications affecting appetite
	Overweight/obesity	BMI >85th percentile for children aged 2–20 years Estimated excessive energy intake Sedentary activity level
	Self-monitoring deficit	Altered self-regulation Overweight/obesity Underweight
Down syndrome A genetic disorder that results from an extra chromosome 21, resulting in developmental problems such as congenital heart disease, cognitive delay, short stature, gastrointestinal problems, and decreased muscle tone.	Inadequate oral intake	Dementia Swallowing difficulty Unintended weight loss
	Breastfeeding difficulty	Feeding difficulty with weak suck Poor weight gain
	Altered GI function	Constipation (related to hypotonia, low activity, and/or low fiber intake) Celiac disease
	Overweight/obesity	Increased body adiposity Estimated excessive energy intake Reduced energy needs related to hypotonia BMI >85th percentile for children aged 2–20 years
Genetic or inherited metabolic disorders Genetic conditions that result in a deficiency or defective enzyme in a metabolic pathway. Some of these, such as phenylketonuria, can result in significant lifetime intellectual and developmental disabilities. Others can cause death if not properly treated.	Imbalance of nutrients Impaired nutrient utilization Altered nutrition-related laboratory values	Food and nutrition knowledge deficit of dietary changes related to new diagnosis Nutrient restriction due to metabolic disorder
	Unintended weight loss	Poor appetite due to metabolic disorder Inadequate energy intake Lack of adequate insurance coverage for low protein foods/special metabolic formulas, limited access to formula Food and nutrition knowledge deficit of dietary changes related to new diagnosis
	Unintended weight gain	Excessive energy intake
	Swallowing difficulty	Abnormal swallow study showing aspiration and/or oral/pharyngeal dysphagia Inadequate oral intake Frequent respiratory infections/pneumonias Coughing/choking with foods/liquids Cleft lip/cleft palate
Orofacial cleft A birth defect that occurs when the lip and/or the roof of the mouth does not form or close properly, resulting in a cleft lip and/or cleft palate.	Breastfeeding difficulty	Inability to form proper latch Inadequate oral intake
	Underweight	Inadequate energy intake Increased energy expenditure

(Continued)

TABLE 45.1 Common Nutrition-Related Issues in Individuals With Intellectual and Developmental Disabilities and Children and Youth With Special Health Care Needs With Suggested Nutrition Diagnosis Terms—Cont'd

Syndrome or Developmental Disability	Nutrition Diagnosis	Etiology and/or Signs and Symptoms
Prader-Willi syndrome A genetic disorder resulting from a deletion in chromosome 15. Hypotonia in infancy can result in failure to thrive before onset of hyperphagia. Common medical issues include obesity as a result of hyperphagia, short stature, and varying levels of intellectual abilities.	Breastfeeding difficulty	Feeding difficulty with weak suck
	Inadequate oral intake	Poor weight gain
	Excessive energy intake	Increased body adiposity
	Overweight/obesity	BMI >85th percentile for children aged 2–20 years Reduced energy needs Self-monitoring deficit
	Limited adherence to nutrition-related recommendations	Inability to limit or refuse foods offered Lack of social support for implementing changes
	Undesirable food choices	Intake inconsistent with diet quality guidelines Unable to independently select foods consistent with food quality and energy-controlled guidelines
	Intake of unsafe food	Food obsession Hyperphagia Eating serves a purpose other than nourishment Pica
Spina bifida (myelomeningocele) A neural tube defect resulting from incomplete closure of the spine during early pregnancy. This results in nerve damage, including neurogenic bowel/bladder and paralysis. Other common medical issues include Arnold-Chiari II malformation, hydrocephalus, growth hormone deficiency, and learning differences.	Increased protein needs	Chronic, nonhealing wounds
	Swallowing difficulty	Abnormal swallow study showing aspiration and/or oral/pharyngeal dysphagia Frequent respiratory infections/pneumonias Coughing/choking with foods/liquids Presence of Arnold-Chiari II malformation of the brain
	Altered GI function	Low fluid and fiber intake Neurogenic bowel Constipation
	Overweight/obesity Unintended weight gain	Increased body adiposity BMI >85th percentile for children aged 2–20 years Estimated excessive energy intake Self-monitoring deficit Limited mobility Reduced energy needs related to altered body composition, short stature

ASD, Autism spectrum disorder; BMI, body mass index; GI, gastrointestinal.

(Adapted from Ptomey LT, Wittenbrook W: Position of the Academy of Nutrition and Dietetics: nutrition services for individuals with intellectual and developmental disabilities and special health care needs, *J Acad Nutr Diet* Apr 115(4):593-608, 2015. doi: 10.1016/j.jand.2015.02.002. PMID: 25819518.).

Nutrition Assessment

Anthropometric Measures

Anthropometric measures are altered when an individual is unable to stand, suffers from contractures or scoliosis, or has other gross motor problems. Measuring body weight may require special equipment such as chair, sling, or bucket scales. Wheelchair scales are used in some clinics but require that the wheelchair weight be known and checked at each clinic visit. Height or length measurements are very challenging to obtain accurately and may be estimated using a recumbent board. Other measures of height include arm span, knee-to-ankle height, or sitting height (see [Fig. 45.1](#) and [Appendix 6](#)).

Although growth charts for children with various syndromes do exist, most clinicians recommend using the general CDC charts (see [Appendix 3](#)) because the information in the specialized charts is often based on small numbers, mixed populations, and old data ([CDC, 2014](#)). For the infant with a developmental disability, the World Health Organization (WHO) growth charts are recommended from birth to 24 months.

Other measures that can be used to explore weight issues include arm circumference, triceps skinfold measures, and body mass index (BMI). BMI is a part of the CDC growth charts and can also be found in [Appendix 3](#). Using the BMI for age can be controversial. For example, BMI is limited for identifying overweight in children with developmental disabilities who are overly fat because of decreased muscle mass and short stature. In addition, BMI may not be accurate if height measurement is not accurate. Height age may also be used to assess BMI if an individual plots far below expected height for age.

Biochemical Measures

Laboratory assessment for the child and adult with developmental disabilities is generally the same as that discussed in [Chapter 5](#) and [Appendix 12](#). Additional tests may be indicated for the individual with epilepsy or seizures who is receiving an anticonvulsant medication such as phenytoin, divalproex sodium, topiramate, or carbamazepine. Use of these medications can lead to low blood levels of folate, carnitine, ascorbic acid, vitamin D, alkaline phosphatase, phosphorus, and pyridoxine.



Fig. 45.1 (A) Knee height measure. (B) Sitting height measure. (C) Arm span measure. (Courtesy Cristine M. Trahms, 2002.)

Assessment of thyroid status is part of the protocol for children with Down syndrome (DS), and a glucose tolerance test is recommended for evaluation of the child with Prader-Willi syndrome (PWS). As appropriate, genetic testing may be encouraged for affected individuals and their biological family members to identify any potential risks to the individual and to future pregnancies.

Dietary Intake and Feeding Problems

Dietary information should be obtained for the child with a developmental disability through a diet history or a food frequency questionnaire (see [Chapter 4](#)). However, obtaining an accurate recall is difficult because written diaries are difficult to obtain when the child has multiple caretakers or when they are in day care or school. When working with an adult with developmental disabilities, it is often difficult to obtain accurate information unless the individual has supervision, such as in special residential housing. Use of pictures and food models is helpful in obtaining an estimate of the individual's intake (see [Box 4.5 in Chapter 4](#)).

Many children and adults with developmental disabilities display feeding problems that seriously impair their ability to eat an adequate diet. Feeding problems are defined as the inability or refusal to eat and/or drink certain foods and liquids because of neuromotor dysfunction, obstructive lesions such as strictures, and behavioral or psychosocial factors. Other causes of feeding problems in this population include

oral-motor difficulties, dysphagia, positioning problems, conflict in parent-child relationships, sensory issues, and tactile resistance from previous intubation ([Heiss et al, 2010](#)). The nutritional consequences of feeding problems include inadequate or excessive weight gain, poor growth in length, lowered immunity, anemia, vitamin and mineral deficiencies, dental caries, and psychosocial problems. Feeding problems should be assessed with an understanding of the anatomy, physiology, and normal development of feeding (see [Chapters 15 to 17 and 41](#)).

Estimates are that feeding problems are found in 40% to 70% of children with special health care needs and 80% of children with developmental delays. Feeding problems are often classified as oral motor, gross motor, behavioral, or sensory, yet feeding problems are usually multifactorial in origin and treatment. **Oral-motor problems** include difficulty with suckling, sucking, swallowing, and chewing. They also include sensory motor integration and problems with self-feeding and are described as exaggerations of normal neuromotor mechanisms that disrupt the rhythm and organization of oral-motor function and interfere with the feeding process ([Box 45.1](#)).

Children with developmental disabilities such as DS, CP, or cleft lip and/or cleft palate (CL/CP) often have oral-motor feeding problems that may be related to the cleft, muscle tone, and inability to accept complex texture. The oral-motor problem also can be related to developmental level, which may be delayed.

BOX 45.1 Oral-Motor Problems

Problem	Description
Tonic bite reflex	Strong jaw closure when teeth and gums are stimulated
Tongue thrust	Forceful and often repetitive protrusion of an often bunched or thick tongue in response to oral stimulation
Jaw thrust	Forceful opening of the jaw to the maximum extent during eating, drinking, attempting to speak, or general excitement
Tongue retraction	Pulling back the tongue within the oral cavity at the presentation of food, spoon, or cup
Lip retraction	Pulling back the lips in a very tight smile like pattern at the approach of the spoon or cup toward the face
Sensory defensiveness	A strong adverse reaction to sensory input (touch, sound, light)

Positioning a child for feeding relates to their motor development, head control, trunk stability, and ability to have hips and legs at a right angle (Figs. 45.2 and 45.3). This is frequently a problem for individuals with CP, spina bifida, and DS. Without proper positioning, oral-motor problems are difficult to correct.

The ability to self-feed may be delayed or absent in the child with developmental disabilities and requires intervention and training by a feeding specialist. A feeding evaluation is best completed with actual observation by a team composed of a speech therapist, a dentist, a physical therapist, an occupational therapist, and an RDN. Frequently, adaptive feeding equipment is needed such as special bottles, nipples, cups, utensils, and so on.

Behavioral issues may result from oral-motor or sensory problems, medical problems, certain medications, and the amount of emphasis placed on feeding. Issues such as control of the feeding process along with lack of autonomy in the child may create negative behavior. Environmental factors also influence the eating behavior of the child (see Chapter 16). Examples include where the child is fed, distractibility, serving sizes, delayed weaning, and frequency of feeding. The feeding relationship concept is important and is further discussed in Chapter 16.

Nutrition Diagnosis

Once the nutrition assessment has been completed, problems should be identified related to growth. Excessive or inadequate weight gain velocity; inadequate or excessive dietary intake; excessive or inadequate fluid intake; altered GI problems such as constipation, gastroesophageal reflux, vomiting, and diarrhea; intake of foods that are unsafe because of contamination or food allergies; food-medication interactions; chewing and swallowing difficulties; and problems with self-feeding may be issues. The nutrition diagnoses should be listed and priorities established, using a family-centered or client-centered approach.

Interventions

Once the nutrition diagnoses are identified and prioritized, short- and long-term goals should be created. Consideration must be given to the motivational level and readiness of the parent or client, their cultural background, and how the therapy can be community based and family centered. This means that consideration must be given to where the client will be served so that it becomes a part of the **individualized education plan (IEP)** or the **individualized family plan**.

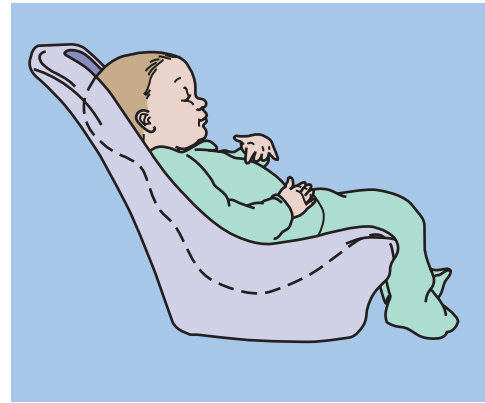


Fig. 45.2 Proper feeding position for the infant. (From Cloud H: *Team approach to pediatric feeding problems*, Chicago, 1987, American Dietetic Association. Reprinted with permission.)

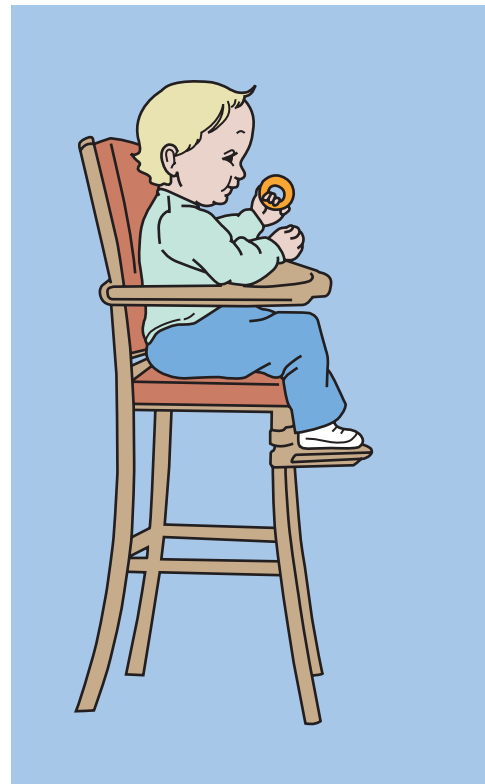


Fig. 45.3 Good feeding position for a child ages 6 to 24 months, showing hip flexion, trunk in midline, and head in midline. Good foot support with a stool should continue throughout childhood. (From Cloud H: *Team approach to pediatric feeding problems*, Chicago, 1987, American Dietetic Association. Reprinted with permission.)

Intervention plans should include all aspects of an individual's treatment program to avoid issuing an isolated set of instructions relevant to only one treatment goal. In some cases MNT may not be the family's first priority in the care of the child or adult, making it important for the RDN to recognize the family's needs and goals (see Chapter 13). Even when the family is ready for an intervention, such as weight management for a child with spina bifida, many factors require consideration. The parent or caregiver's educational level and income,

language barriers, access to safe and appropriate food, and family coping strategies always should be identified (see [Chapter 13](#)).

Monitoring and Evaluation

Once MNT has been initiated, the need for follow-up evaluation and monitoring by either the RDN or another health care professional is important. Giving information in writing, ideally followed by a phone call, gives the chance to repeat some of the discussion and to answer any questions not asked during the initial session. Clarification of suggestions often is needed when monitoring nutrition changes that affect growth and development; a follow-up visit also may be needed.

A case manager may be involved who communicates with the adult with the disability, and a family resource coordinator may be available to assist families of children with disabilities. The RDN may need to assist in finding appropriate resources to pay for supplemental nutrition products, tube feedings, and special food products as a part of the follow-up process. Community and agency resources are identified and discussed, such as the Special Supplemental Nutrition Program for Women, Infants, and Children (WIC) and other government and non-government assistance programs.

CHROMOSOMAL ABNORMALITIES

Down Syndrome

Down syndrome (DS) is a chromosomal aberration of chromosome 21 (trisomy 21). It has an incidence of 1 in 600 to 800 live births and results from the presence of an extra chromosome in each cell of the body. This anomaly causes the physical and developmental features of short stature; congenital heart disease; intellectual disability; decreased muscle tone; hyperflexibility of joints; speckling of the iris (Brushfield spots); upward slant of the eyes; epicanthal folds; small oral cavity; short, broad hands with the single palmar crease; and a wide gap between the first and second toes ([Bull and Committee on Genetics, 2011](#); [Fig. 45.4](#)).

Pathophysiology

Normally every cell of the human body except for the gametes (sperm or ova) contains 46 chromosomes, which are arranged in pairs (see



Fig. 45.4 Child with Down syndrome. (From www.istockphoto.com.)

[Chapter 6](#)). With DS there is one extra chromosome for a total of 47. This anomaly can occur by one of these three processes: nondysjunction, translocation, and mosaicism. In **nondisjunction**, chromosome 21 fails to separate before conception and the abnormal gamete joins with a normal gamete at conception to form a fertilized egg with three of chromosome 21. This also may occur during the first cell division after conception. This type of DS is usually sporadic and has a recurrence rate of 0.5% to 1%. In **translocation**, the extra chromosome is attached to another chromosome (usually 14, 15, or 22). Approximately half the time, this type of DS is inherited from a parent who is a carrier; it has a much higher risk of recurrence in a subsequent pregnancy. In **mosaicism** the abnormal separation of chromosome 21 occurs sometime after conception. All future divisions of the affected cell result in cells with an extra chromosome. Therefore the child has some cells with the normal number of chromosomes and other cells with an extra chromosome. Frequently the child with this type of DS lacks some of the more distinctive features of the syndrome ([Bull and Committee on Genetics, 2011](#)) (see *Pathophysiology and Care Management Algorithm: Down Syndrome*).

Medical Treatment

The National Down Syndrome Congress has published a listing of the health concerns for individuals with DS; many have nutrition implications ([Table 45.2](#)). The American Academy of Pediatrics (AAP) has published health supervision guidelines for children and young adults up to age 21 ([Bull and Committee on Genetics, 2011](#)).

Medical Nutrition Therapy

Anthropometric measures. Height, weight, head circumference, triceps skinfold, and arm circumference are obtained for the child with

TABLE 45.2 Medical Problems Common in Down Syndrome

Condition	%
Hearing problems	75
Vision problems	60
Cataracts	15
Refractive errors	50
Obstructive sleep apnea	50–75
Otitis media	50–70
Congenital heart disease	40–50
Hypodontia and delayed dental eruption	23
Gastrointestinal atresia	12
Thyroid disease	4–18
Anemia	3
Iron deficiency	10
Transient myeloproliferative disorder	10
Leukemia	1
Celiac disease	5
Atlantoaxial instability	1–2
Autism spectrum disorder	1
Hirschsprung disease	<1

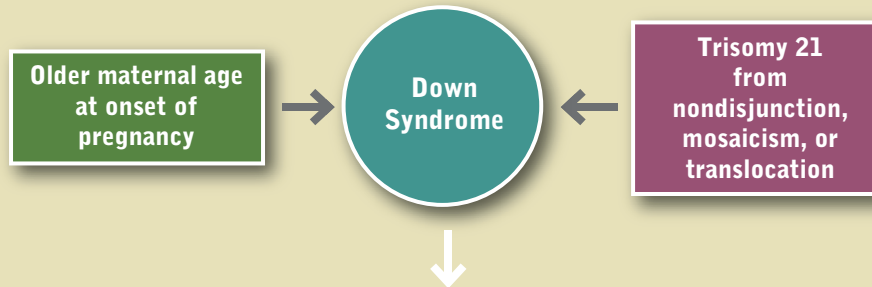
(From Bull MJ: Committee on genetics: health supervision for children with Down syndrome, *Pediatrics* 128:393–406, 2011.)



PATHOPHYSIOLOGY AND CARE MANAGEMENT ALGORITHM

Down Syndrome

ETIOLOGY



PATHOPHYSIOLOGY

Clinical Findings

- Hypotonia
- Hyperflexibility/mobility of the joints
- Hip subluxation or dislocation
- Scoliosis
- Foot deformity
- Microbrachycephaly
- Short neck
- Depressed nasal bridge and small nose
- Upward-slanting eyes
- Abnormally shaped ears
- Enlarged, protruding tongue
- Single simian crease in center of the palm
- Excessive space between large and second toe
- Mental retardation
- Speech and motor delay
- Cardiac anomalies
- GI atresia or stenosis
- Hearing loss
- Hypothyroidism
- Dental problems
- Cataracts

MANAGEMENT

Nutrition Assessment

- Feeding problems
- Nutritional intake
- Fluid intake
- BMI compared with Down syndrome standards
- Weight changes
- Dysphagia
- Hemoglobin concentration

Medical Management

- Monitor for leukemia
- Management of infections
- Management of respiratory problems
- Signs of Alzheimer's disease later
- Annual monitoring of hypothyroidism

Nutrition Care

- Assess for feeding challenges
- Provide supportive feeding environment
- Use of adaptive feeding utensils
- Monitor weight changes
- Monitor for signs of nutrient deficiencies

DS with the usual measurements (see [Chapter 5](#)). BMI can be taken but may be higher than normal because of short stature. Muscle tone is low and gross motor ability is often delayed, leading to the possibility of the individual becoming overweight. Monitoring should be frequent, and growth is plotted on the CDC charts or the WHO charts for the younger child (see [Appendix 3](#)).

Biochemical measures. Numerous studies have shown biochemical and metabolic abnormalities in individuals with DS; however, many have involved small samples and were difficult to interpret ([Bull and Committee on Genetics, 2011](#)). Although serum concentrations of albumin have been found to be low, the guidelines from the Down Syndrome Medical Congress do not list serum albumin assessment as routine. Increased glucose levels have been reported, with an increased incidence of diabetes mellitus.

Current guidelines for the treatment of infants and children with DS include evaluation of thyroid function at birth and thereafter annually. A number of studies have looked at zinc, copper, and selenium as possible deficiencies but concluded that they do not usually exist in children with DS as long as food access and appetite are good ([Lima et al, 2010](#)).

Dietary intake. During infancy, the food intake of the infant with DS may differ from that of the typically developing infant. Although human breastmilk is recommended, many infants with DS are formula fed. Infant illnesses, admission to the neonatal unit, frustration, depression, perceived milk insufficiency, and difficulty in suckling by the infant are some of the reasons why formula feeding is used.

Progression to solid food has been found to be delayed in children with DS, mostly as a result of delays of oral-motor development. Introduction of solid food may not be offered at 6 months if the infant has poor head control or is not yet sitting. Low tone and sucking problems also delay weaning from the breast or bottle to the cup. IEPs include feeding and feeding progression instruction and practice.

An important part of evaluating dietary intake is determining energy and fluid needs, because children with DS have a high prevalence of obesity. Studies have indicated that the resting energy expenditure (REE) of the child with DS is lower than for children without DS—and may be as much as 10% to 15% lower than the dietary reference intake (DRI) for energy ([Medlen, 2006](#)). For the child older than age 5, calculations for energy requirements may have to be based on height rather than weight ([Table 45.3](#) and see [Chapter 2](#)).

Feeding skills. Feeding skills are delayed in the infant and child with DS. Some parents find it difficult to initiate oral-motor skills such as suckling and sucking. The infant with DS often has difficulty in coordinating sucking, swallowing, and breathing, which are the foundations for early feeding. When the infant has a congenital heart defect, which occurs in 40% to 60% of DS infants, sucking is weakened, and fatigue interferes with the feeding process. GI anomalies are found in 8% to 12% of infants with DS, and these infants often require nasogastric or gastrostomy feedings.

Other physical factors that make feeding difficult in the first years of life include a [midfacial hypoplasia](#) (a craniofacial deformity common in cleft palate), a small oral cavity, a small mandible, delayed or abnormal dentition, misaligned teeth, nasal congestion, facial hypotonia, and small hands and short fingers. Weaning and self-feeding are usually late compared with the typically developing infant and frequently do not emerge until 15 to 18 months of age. The DS infant strives for independence and autonomy approximately 6 months later than the child without DS.

Intervention Strategies

Overweight. Children with DS (and all children who have low muscle tone) are at increased risk for becoming overweight because of

TABLE 45.3 Estimated Caloric Needs for Special Conditions

Condition	kcal/cm of height	Comments
Developmentally normal child	Average 16	
Prader-Willi syndrome	Maintain growth: 10–11 Promote weight loss: 8.5	For all children and adolescents
Cerebral palsy		
Mild	14	For children ages 5–11 years
Severe, limited mobility	11	For children ages 5–11 years
Down syndrome	Girls: 14.3 Boys: 16.1	For children ages 5–11 years
Motor dysfunction		
Nonambulatory	7–11	For children ages 5–11 years
Ambulatory	14	For children ages 5–11 years
Spina bifida	Maintain weight: 9–11 Promote weight loss: 7	For all children older than 8 years of age and minimally active

(Modified from Weston S, Murray P: Alternative methods of estimating daily energy requirements based on health condition. In Devore J, Shotton A, editors: *Pocket guide to children with special health care and nutritional needs*, Chicago, 2012, Academy of Nutrition and Dietetics.)

lower than typical energy expenditure. Weight management includes assessing the feeding developmental level of the child, their physical capability related to gross motor skills and physical activity levels, and environmental challenges. Environmental recommendations are the same as those for all children and include following a regular eating schedule that includes three meals and two to three snacks at regular times with the child sitting either in a high chair or at the table. Planned snacks should be nutrient dense, avoiding foods with excessive amounts of fat and sugar. Soft drinks should be discouraged and physical activity encouraged. It may be helpful to design a calorie-controlled eating plan based on kilocalories per centimeter of height as shown in [Table 45.3](#) ([Murray and Ryan-Krause, 2010](#)).

Counseling in which the parent or caregiver helps to determine a realistic plan that emphasizes appetite awareness and self-regulation may focus on serving sizes, healthy food choices, and food preparation and decreasing the number of times meals are purchased in fast-food restaurants. If the child or adolescent is school age, a prescription for a special meal at school can be obtained by using the school food service prescription (to be discussed later in the chapter). Prevention of overweight and promotion of an active lifestyle are the goals. Recent studies involving adolescents and young adults have addressed the success of exercise and healthy lifestyles for obese participants with DS. Successful programs include parent training in behavioral intervention and nutrition and activity education over multiple sessions ([Curtin et al, 2013](#)).

Eating skills. Parents may wrongly expect more typical eating development for the child with DS. An example of this is the unnecessary delay of weaning to a cup or progression of food textures because of inadequate effort or education. Behavioral problems related to

feeding may develop based on what happens between the parent and child at mealtime. An intervention program with a feeding team can guide the parent in working with the child toward attainable feeding skills related to the developmental level.

Constipation. This is a frequent problem for the child with DS because of overall low tone that is exacerbated by a lack of fiber and fluid in the diet. Treatment should involve providing adequate fiber and fluid, with water consumption emphasized. Fiber content of the diet for children after age 3 is 5 to 6 g per year of age per day. For adults the recommendation is for 25 to 30 g of dietary fiber daily (see Chapter 28).

Prader-Willi Syndrome

Prader-Willi syndrome (PWS) was first described in 1956 by Prader, Willi, and Lambert. It is a genetic condition caused by the absence of chromosomal material. PWS occurs with a frequency of 1 in 10,000 to 1 in 25,000 live births. Characteristics of the syndrome include developmental delays, poor muscle tone, short stature, small hands and feet, incomplete sexual development, and unique facial features. Insatiable appetite leading to obesity is the classic feature of PWS; however, in infancy the problem of **hypotonia** (low muscle tone) interferes with feeding and leads to failure to thrive (Miller et al, 2011; Fig. 45.5). Developmental delays (affecting 50% of the population), learning disabilities, and intellectual disabilities (affecting 10%) are associated with PWS.

The genetic basis of PWS is complex. Individuals with PWS have a portion of genetic material deleted from chromosome 15 received from the father. Of the cases of PWS, 70% are caused from the paternal deletion, occurring in a specific region on the q arm of the chromosome. PWS also can develop if a child receives both chromosome 15s from the mother. This is seen in approximately 25% of the cases of PWS and is called maternal uniparental disomy. Early detection of PWS is now possible because of the use of DNA methylation analysis, which correctly diagnoses 99% of the cases (Miller et al, 2011). This is an important development in the early identification and subsequent treatment of these children to prevent obesity and growth restriction, and it is used to identify the infant born with features and characteristics described previously.

Pathophysiology

Metabolic abnormalities. Short stature in the individual with PWS has been attributed to growth hormone (GH) deficiency. In addition

to decreased GH release, children have low serum insulin-like growth factor (IGF)-1, low IGF-binding protein-1, and low insulin compared with developmentally typical obese children. GH therapy was approved by the Food and Drug Administration (FDA) in 2000, and in one 5-year study in Japan, 37 patients from age 3 to 21 years experienced significant increase in height gain velocity when given GH (Obata et al, 2003). A more recent study found that GH therapy of infants and toddlers for 12 months significantly improved body composition and mobility skill acquisition (Carrel et al, 2010).

In addition to the GH deficiency, individuals have a deficiency in the hypothalamic-pituitary-gonadal axis, causing delayed and incomplete sexual development. Finally, there is a decreased insulin response to a glucose load in children with PWS compared with age-matched non-PWS obese children (Haqq et al, 2011).

Appetite and obesity. Appetite control and obesity are common problems for individuals with PWS. After the initial period of failure to thrive, children begin to gain excessively between the ages of 1 and 4, and appetite slowly becomes excessive. Based on longitudinal study, Miller et al, (2011) describe this gradual and complex progression in terms of seven nutritional phases based on levels of appetite, metabolic changes, and growth. In fact, some adults with PWS may progress to the last phase, with no insatiable appetite, and the person is able to feel full.

This uncontrollable appetite, a classic feature of PWS, when combined with overeating, a low basal metabolic rate, and decreased activity, leads to the characteristic obesity. The cause of the uncontrollable appetite is suspected to involve the hypothalamus and altered levels of satiety hormones and peptides such as ghrelin (Scerif et al, 2011).

Body composition is an important consideration in the evaluation of individuals with PWS. They have decreased lean body mass and increased body fat, even in infancy (Reus et al, 2011). Body fat generally is deposited in the thighs, buttocks, and abdominal area. The lowered energy expenditure is found in young children, adolescents, and adults with PWS, with one study showing adolescents with PWS having a total energy expenditure (TEE) 53% of that of typically developing adolescents who have a larger body size (McCune and Driscoll, 2005). The low muscle tone contributes greatly to the lack of interest in physical activity.

Nutrition Assessment

Anthropometric measures. Height measurements tend to be lower in PWS infants and young children, with the rate of height gain tapering off between the ages of 1 and 4. The usual measurements of length or height, weight, and head circumference should be taken and plotted on the CDC growth curves. Other clinically relevant measurements include arm circumference and triceps skinfold. BMI may be distorted for the individual with PWS because of the short stature; however, plotting the BMI over time is useful in determining unusual changes (see Appendix 3). It is important that anthropometric measures be taken frequently and reported to the parents or caregiver.

Biochemical measures. Biochemical studies are generally the same for the PWS individual, with the exception of either fasting blood glucose tests or glucose tolerance tests. These are added because of the risk for diabetes mellitus, possibly related to the decreased insulin response and obesity that usually accompany PWS.

Dietary intake. Dietary information varies for individuals with PWS depending on their age. In infancy the dietary information should be obtained with a careful dietary history and analyzed for energy and nutrient intake. Infants are commonly difficult to feed because of their hypotonia, poor suck, and delayed motor skills. In general, their feeding development is slower than in the typically developing infant, and transitioning to food at 4 to 6 months of age may be difficult. Many



Fig. 45.5 Child with Prader-Willi syndrome.

infants with PWS have gastroesophageal reflux, requiring medication or thickening of their formula. Some infants may require nasogastric or G-tube feeds initially. A video fluoroscopic swallow study (VFSS) may be recommended to rule out dysphagia and aspiration. VFSS is an x-ray done using barium, and the speech or occupational therapist determines whether there is any aspiration on various liquids. Feeding therapy is also beneficial. Our clinic has been requesting VFSS on infants, and we have been finding silent aspiration. We have been proactive in including speech and occupational therapy as part of the assessment (Salehi et al, 2017).

During the toddler years, weight gain may increase rapidly as dietary intake increases. This requires careful assessment of portion sizes, frequency of feeding, and types of foods served. Although some parents may report that the child with PWS does not eat more than other children in the family, they have to be educated that the energy needs of the child are lower because of the reduced lean muscle mass and slow development of motor skills and activity. As the child gets older, interest in food increases, and, starting around ages 5 through 12, the child may be hungry all the time and display difficult behaviors such as tantrums, stubbornness, and food stealing. Information gathered during the dietary interview should include asking about feeding routine and management. The phrase “No doubt, no hope (no chance), no disappointment” has been used. With routines there is no doubt the child is fed and no chance of getting extra foods, leading to no disappointment. Determination of energy needs for the infant with PWS is the same as for a developmentally typical infant. However, when entering the toddler years, children with PWS will need fewer calories to maintain weight gain along the growth curve. This will apply in adulthood when fewer calories are needed to maintain weight. Energy needs have been calculated according to centimeters of height from 2 years and older (Miller et al, 2013). It has been recommended that the macronutrient intake of the diet be 25% protein, 45% carbohydrate, and 30% fat and at least 20 g of fiber. A reduced energy intake and well-balanced diet improve weight control in children with PWS (Miller et al, 2013) (see Table 45.3).

Feeding skills. The infant with PWS often presents with weak oral skills and poor sucking skills in the first year of life. As the child matures, feeding skills are not a problem, but they may be delayed. Chewing and swallowing problems usually are not seen, although they may be associated with the low muscle tone. Behavioral feeding issues are associated with an insatiable appetite and not being provided with food. This can bring about tantrums.

Intervention Strategies

Intervention for PWS should occur at each developmental stage: infancy, toddler, preschool age, school age, and adult.

Infancy. Providing adequate nutrition as established by the AAP related to breastfeeding or formula feeding is recommended. Because feeding may be difficult related to sucking, concentrating the formula or breastmilk may be necessary to promote adequate weight gain. Feeding intervention will assist in improving the sucking problems caused by hypotonia. As the infant matures, a concentrated formula is not necessary, and solid foods can be added when head control and trunk stability are achieved, usually at approximately ages 4 to 6 months.

Toddler and preschool age. Most children begin to gain excessive weight between 1 and 4 years of age. Beginning a structured dietary plan for the child and the family is important so that the toddler learns that meals are provided at specified times and a pattern of grazing does not develop. Encourage parents and caregivers to provide balanced meals including protein, vegetables, whole grains, and fruits and limited amounts of sweets. Early intervention during the preschool years is very important in working with feeding issues and intake regulation

as children grow older. Monitoring weight, height, and nutrient intake monthly will help to identify early signs of weight acceleration so that dietary patterns and meal composition can be assessed and modified as needed. Regular physical activity is an important part of the IEP, and physical therapy services can be made available if necessary.

School age. For the school-age child, collaboration with the school food service program becomes important. Energy needs should be calculated per centimeter of height (see Table 45.3) and are generally 50% to 75% of the energy needs of typically developing children. This may require using a prescription for special meals through the school food service program.

At home, environmental controls may be required, with cupboards and even kitchens being locked, because the child and adolescent have limited satiety and will search for food away from mealtime. Some parents report that GH therapy for their child helps, but it does not seem to change the child's lack of satiety. Appetite-suppressing medications have been used but are largely unsuccessful.

Adulthood. Prevention of obesity is truly the key for successful treatment of PWS; however, many adults who are not identified early become very obese. Weight management programs providing a very low 6 to 8 kcal per centimeter of height may be required. If a lower-calorie diet is being recommended, patients will likely benefit from vitamin-mineral and essential fatty acid (EFA) supplements if micronutrient needs cannot be met through the diet. Many dietary treatments have been tried such as the ketogenic diet (see Appendix 19) and the protein-sparing modified-fast diets. However, with any approach, strict supervision is usually required, and great emphasis must be placed on physical activity. A behavior management approach also has been recommended to implement the dietary management and physical activity plans. In many states there are group homes for adults with PWS, in which supervised independent living is possible and meals can be very structured and exercise programs implemented.

MNT of children and adults with PWS requires follow-up with many health care providers and schools. Fortunately parents of the individual with PWS now have access to a number of support groups and organizations dedicated to education, research, and establishing treatment programs.

NEUROLOGIC DISORDERS

Spina Bifida

Spina bifida is a neural tube defect (NTD) that presents in a number of ways: meningocele, **myelomeningocele (MM)**, and spina bifida occulta. MM is the most common derangement in the formation of the spinal cord and generally occurs between 26 and 30 days of gestation, with the date of occurrence affecting the location of the lesion. The lesion may occur in the thoracic, lumbar, or sacral area and influences the amount of paralysis. The higher the lesion, the greater is the paralysis. Manifestations range from weakness in the lower extremities to complete paralysis and loss of sensation. Other manifestations include incontinence and hydrocephalus (fluid accumulation in the brain). The incidence of spina bifida is approximately 2.7 to 3.8 per 10,000 live births in the United States per year, with Hispanics having the highest incidence (Canfield et al, 2014).

Prevention of spina bifida is now possible. In the 1980s, studies reported a positive effect from supplementation of mothers with folic acid plus multivitamins (Smithells et al, 1983). This reduced the risk of a second pregnancy with spina bifida as an outcome. As a result of numerous studies showing folic acid supplementation before conception to be effective, the national recommendation is 400-800 mcg/day for all women of childbearing age, and 4 mg for women who have had

previous affected pregnancies (1). Through its Public Health Program, the WHO advocates globally for folic acid supplementation. Folic acid has been added to many flours and other cereal and grain products in the food supply since 1996. The CDC has reported that mandatory folic acid fortification of cereal and grain products has helped approximately 1300 babies to be born without NTDs every year (Centers for Disease Control and Prevention (CDC), 2015). These public health measures have resulted in increased folate blood levels in US women of childbearing age and a decrease of 20% in the national rate of spina bifida (Das et al, 2013; see Chapter 14).

Pathophysiology

The spinal lesion may be open and is typically repaired surgically shortly after birth, usually within 24 hours, to prevent infection. Although the spinal opening can be repaired surgically, the nerve damage is permanent, resulting in the varying degrees of paralysis of the lower limbs. In addition to physical and mobility issues, most individuals have some form of learning disability.

The spinal lesion affects many systems of the body and can result in weakness in the lower extremities, paralysis, and nonambulation; poor skin condition resulting in pressure injuries (see Chapter 19); loss of sensation with bowel and bladder incontinence; hydrocephalus; urinary tract infections; constipation; and obesity. Seizures also occur in approximately 20% of children with MM and may require medication. If medication fails, a ketogenic diet can be tried in some cases (see Appendix 19). Chronic medication is frequently required for prevention and treatment of urinary tract infections and for bladder control. The resultant nutrition problems include obesity, feeding problems, constipation, and drug-nutrient interactions. According to the Spina Bifida Association, children with spina bifida have an increased risk of developing an allergy to latex because of repeated exposures to latex products. The risk is decreasing with the use of latex-free products. Children who are allergic to latex may also develop an allergy to certain foods that cross-react with latex such as banana, avocado, chestnut, kiwi, apple, carrot, celery, papaya, tomato, and melon. The Spina Bifida Association recommends that these foods should not be avoided unless there is an established reaction.

Nutrition Assessment

Anthropometric measures. Infants and children with spina bifida are usually shorter than typically developing children because of reduced length and atrophy of the lower extremities, although other problems such as hydrocephalus, scoliosis, renal disease, and malnutrition may contribute. The level of the lesions also can affect the length and height of the individual.

Obtaining accurate length and height measures can be difficult, especially as the child grows older. An alternate measure for determining height, the arm span/height ratio, is used and modified, depending on leg muscle mass. Arm span can be used directly as a height measure ($\text{arm span} \times 1$) if there is no leg muscle mass loss, as in a sacral lesion. $\text{Arm span} \times 0.95$ can be used to determine height if there is partial leg muscle loss, and $\text{arm span} \times 0.90$ is used for a height measurement when there is complete leg muscle loss, such as with a thoracic spinal lesion (Kreutzer and Wittenbrook, 2013; see Fig. 45.1 and Appendix 6).

Weight measures can be obtained for the child unable to stand by using chair, sling, bucket, or wheelchair scales. To monitor weight accurately, it should be obtained in a consistent manner, with the person in light clothing or, if younger than age 2 years, undressed and wheelchairs weighed at each visit. Triceps skinfold measures can also be used, along with subscapular measures and abdominal and thorax measures, to determine the amount of body fat. They may be used as adjuncts to stature and weight measurements.

Head circumference should be measured in infants and toddlers up to age 3. A high percentage of children with spina bifida have head shunts as a result of their hydrocephalus. Unusual changes in the size of the head may indicate a problem with the shunt.

Biochemical measures. Useful biochemical tests in the health monitoring of individuals with spina bifida include iron status tests, measurements of vitamin D, and other tests related to the nutritional consequences of medications needed for seizures and urinary tract infection control (see Chapter 5 and Appendix 12).

Dietary intake. Many children with spina bifida eat a limited variety of foods, and they are frequently described as “picky eaters” by the parents. When doing a dietary history, it is important to ask about the variety of foods, particularly of high-fiber foods. The school-age child may be prone to skipping breakfast because early morning preparations for school require more time than for the nonaffected child.

Energy needs are lower for the child with spina bifida (see Table 45.3), and calorie requirements must be determined carefully to prevent the obesity to which many are prone. Ekvall and Cerniglia (2005) found that for MM children 8 years or older, the caloric need is 7 cal/cm of height for weight loss and 9 to 11 cal/cm of height to maintain weight. They also found that it is important to evaluate how the mother or caretaker perceives food for the child, because it represents sympathy and love for many parents.

It is important to evaluate fluid intake because so many children are prone to having urinary tract infections as a result of daily catheterization. They may be drinking inadequate amounts of water and excessive amounts of soft drinks or tea that contain caffeine, which is a mild diuretic. Cranberry juice can be offered because it may help to prevent urinary tract infections, but calorie intake must be taken into consideration. Physical activity needs to be evaluated and may be found to be very limited, particularly when the child is nonambulatory. Ambulatory individuals with a shunt may be restricted from contact sports but can be involved in walking and running.

Feeding skills need to be evaluated, along with oral-motor function in particular. Many children with spina bifida are born with **Arnold-Chiari malformation of the brain**, which affects the brainstem and swallowing (see Chapter 41 and Appendix 20 for dietary recommendations for dysphagia). Difficulty in swallowing may contribute to the child avoiding certain foods. Because of this there may be delays in weaning from the breast or bottle to the cup, but there should be no delays in gaining self-feeding skills.

Clinical evaluation. Evaluation should include looking for pressure injuries and signs of dehydration, along with asking about the amount and type of foods and fluids consumed. Constipation may be caused by the neurogenic bowel (fecal incontinence due to lack of normal bowel function) combined with a diet low in fiber and fluids.

Intervention Strategies

Many children with spina bifida are overweight. The Spina Bifida Association reports that after age 6, 50% of children are overweight and in adolescence and adulthood more than 50% are obese (Spina Bifida Association, 2017). This may occur when ambulation is a problem, leading to decreased energy needs. Refusal to accept a wide variety of foods is also common. Frequent feeding may be due to an oral-motor or a behavioral problem. Counseling includes introducing foods around age 6 months, limiting the intake of high-sucrose infant jar foods, and supporting the child to accept a wide variety of flavors and textures.

In addition to the generally known health consequences of obesity, specific concerns for individuals who have spina bifida include further limits of mobility and increased pressure on skin, increasing the already high risk of skin breakdown. Increased social isolation and decreased self-worth are also consequences (Spina Bifida Association, 2017).

Obesity prevention includes addressing the problems of limited physical activity, increasing fluids and fiber, and estimating the appropriate amount of calories. For school-age children it may be useful to provide a prescription for a low-calorie breakfast and lunch, with weight management listed as a part of the IEP. Enrollment in a group weight management program has been used successfully along with encouraging physical activity. The ideal program uses a team approach with involvement of the individual, the family, RDN, nurse, occupational therapist, physical therapist, educator, and psychologist (see Chapter 21).

In many clinics the child or adult with spina bifida is seen on a semiannual or annual basis. This frequent follow-up is necessary and should include monitoring of growth, particularly weight; food and fluid intake; and medication use. School programs and IEPs are excellent follow-up tools; however, the school often lacks appropriate scales for weighing a nonambulatory student. In this situation the parent should be encouraged to bring the child to the clinic for weight checks or, if distance is a problem, find a long-term care facility that will permit use of its scales. Follow-up by phone contact or e-mail can be done for evaluating dietary intake and fluid management.

Cerebral Palsy

Cerebral palsy (CP) is a group of disorders of motor control or coordination resulting from injury to the brain during its early development. Among the causative agents of CP are prematurity; blood-type incompatibility; placental insufficiency; maternal infection that includes German measles; other viral diseases; neonatal jaundice; anoxia at birth; and other bacterial infections of the mother, fetus, or infant that affect the central nervous system.

The problem in CP lies in the inability of the brain to control the muscles, even though the muscles themselves and the nerves connecting them to the spinal cord are normal. The extent and location of the brain injury determine the type and distribution of CP. The incidence of CP varies with different studies, but the most commonly used rate is 1.5 to 4 in 1000 live births. The prevalence of premature births has contributed to maintenance of this figure despite electronic fetal monitoring (Fig. 45.6).

Pathophysiology

There are various types of CP, which are classified according to the neurologic signs involving muscle tone and abnormal motor patterns and postures. The diagnosis of CP is generally made between 9 and 12 months of age and as late as 2 years with some types (Box 45.2). The severity of CP is classified by levels of self-sufficiency in activities of

daily living (ADLs). There are five categories of gross motor function, with category one being the least affected and five the most affected (Box 45.3).

Poor nutrition status and growth failure, often related to feeding problems, are common in children with CP. Meeting energy and nutrient needs is particularly difficult in children and adults with more severe forms of CP such as spastic quadriplegia and athetoid CP, which is typically classified as Gross Motor Function Classification System (GMFCS) 4 to 5.

For example, bone mineral density of children and adolescents with moderate to severe CP is reduced in those with gross motor function and feeding difficulties (Andrew and Sullivan, 2010).

BOX 45.2 Different Types of Cerebral Palsy

Spastic cerebral palsy (CP): Spasticity implies increased muscle tone. This type accounts for 70%–80% of cases. Muscles continually contract, making limbs stiff, rigid, and resistant to flexing or relaxing. Reflexes can be exaggerated, while movements tend to be jerky and awkward. Often, the arms and legs are affected. The tongue, mouth, and pharynx can be affected as well, impairing speech, eating, breathing, and swallowing.

Nonspastic CP: Ataxic CP affects coordinated movements. Balance and posture are involved. Walking gait is often very wide and sometimes irregular. Control of eye movements and depth perception can be impaired. Often, fine motor skills requiring coordination of the eyes and hands, such as writing, are difficult. Does not produce involuntary movements, but instead indicates impaired balance and coordination.

Dyskinetic: Dyskinetic CP is separated further into two different groups; athetoid and dystonic.

Athetoid CP includes cases with involuntary movement, especially in the arms, legs, and hands.

Dystonia/Dystonic CP encompasses cases that affect the trunk muscles more than the limbs and results in fixed, twisted posture.

Because nonspastic CP is predominantly associated with involuntary movements, some may classify CP by the specific movement dysfunction, such as:

Athetosis—slow, writhing movements that are often repetitive, sinuous, and rhythmic

Chorea—irregular movements that are not repetitive or rhythmic, and tend to be more jerky and shaky

Choreoathetoid—a combination of chorea and athetosis; movements are irregular but twisting and curving

Dystonia—involuntary movements accompanied by an abnormal, sustained posture

Mixed CP: A child's impairments can fall into both categories, spastic and nonspastic, referred to as mixed CP. The most common form of mixed CP involves some limbs affected by spasticity and others by athetosis.



Fig. 45.6 Child with cerebral palsy. (From www.istockphoto.com.)

BOX 45.3 Gross Motor Function Classification System

The 5-level gross motor function classification system (GMFCS) is used in clinical and research settings.

1. Walks without limitations
2. Walks with limitations
3. Walks using a hand held mobility device
4. Self-mobility with limitations, may use powered mobility
5. Transported in a manual wheelchair
 - fed orally without a feeding tube
 - fed with a feeding tube

Other health problems include constipation exacerbated by inactivity, abnormal muscle coordination, and lack of fiber and fluids. Dental problems occur and often are related to malocclusion, dental irregularities, and fractured teeth. Lengthy and prolonged bottle-feedings of milk and juice promote the decay of the primary upper front teeth and molars (see [Chapter 25](#)). Hearing problems and especially visual impairments, intellectual disability, respiratory problems, and seizures affect nutrition status. Seizures are controlled with anticonvulsants, and a number of drug-nutrient interaction problems occur (see [Appendix 13](#)).

Nutrition Assessment

Anthropometric measures. This is an important area of assessment because of growth failure in the more severely involved child or adult with CP. Children with CP are often shorter and, depending on the level of severity, some children with CP may have to be measured for length using alternative measuring techniques such as recumbent length boards or standing boards as they grow older (see [Appendix 5](#)). However, some of the measuring devices are inappropriate for the child with contractures and inability to be stretched out full length. Arm span can be used when the individual's arms are stretchable, as well as upper arm and lower leg length. [Stevenson \(1995\)](#) has recommended lower leg length or knee height as a possible measure for determining height for children and adults with lower-leg CP (see [Fig. 45.1](#)). The CDC recommends using the CDC/WHO curves designed for nonaffected children, plotting sequentially for indications of malnutrition rather than using the disease-specific curves.

Weight measures should be collected over time. Scales may require modifications, with positioning devices for the individual with CP who has developed scoliosis, contractures, and spasticity. Working with a physical therapist to find a positioning device that can be placed in a chair scale or using a bucket scale often works well. Midupper arm circumference and triceps skinfold measures are recommended reliable ways to screen for fat stores in children with CP. Head circumference should be measured regularly from birth to 36 months and plotted on the CDC growth charts (see [Appendix 3](#)).

Biochemical measures. Although no specific laboratory tests are indicated for the child with CP, a complete blood count, including hemoglobin, hematocrit, and ferritin, should be done when food intake is limited and malnutrition is a possibility. Because bone fractures are a significant problem for many children and adults with spastic quadriplegia, bone mineral density may require evaluating. Medications for seizures may be given; many have nutrition interaction problems (see [Appendix 13](#)). Evaluation of vitamin D, carnitine, and vitamin K levels may be useful.

Dietary intake. Oral-motor difficulties can result in limiting the intake of food and fluid, making it difficult for caretakers to meet nutritional needs. The energy needs of the individual with CP vary according to the type of CP. Studies show that the REE and TEE are lower in those with spastic quadriplegic CP than in developmentally typical controls (see [Table 45.3](#)).

Intervention Strategies

A high percentage of children with CP have feeding problems that are largely the result of oral-motor, positioning, and behavioral factors. As infants they may have difficulty swallowing and after infancy, coordinating chewing and swallowing, so the normal progression to solid foods is later than usual. Oral-motor problems may be permanent, requiring lifelong modifications such as modified textures of foods and fluids to ensure swallowing safety and the need for tube feeding for part or all of nutrition and hydration needs. All this may lead to inadequate intake and growth limitations. For those infants and children with IEPs, the team of RDN, speech therapist, occupational therapist,

and physical therapist should evaluate the problem and work together in planning therapy to maximize skill development.

Gastroesophageal reflux frequently is seen in these infants and toddlers. Tube feeding may be required if swallowing studies reveal aspiration. Alternative techniques in feeding should be considered, which could include modifying food textures, thickening all beverages, or placing a gastrostomy tube ([Mahant et al, 2011](#)). RDNs should evaluate gastrostomy feedings for caloric and nutritional value and volume required for hydration as well as including oral solids and liquids into the tube-feeding formula if possible.

Typical problems identified in the evaluation are altered growth, inadequate energy or fluid intakes, drug-nutrient interactions, constipation, and feeding problems. Working out an intervention plan is most successful when it involves the parent as part of the team, addresses cultural issues, and recognizes the importance of the feeding problem. Children with CP have complex problems that require routine follow-up with the family and appropriate agencies, schools, and institutions in the community. State agencies often provide tube-feeding formulas and special wheelchairs and equipment to assist with feeding problems. These agencies vary from state to state.

Autism

Autism spectrum disorders (ASDs) were originally described in the 1940s as a behavioral disorder. It is now understood to be a pervasive and systemic syndrome with neurologic, immunologic, GI, and endocrine involvement. There is no test to diagnose ASD at this time; rather ASD is diagnosed by the presence of impaired reciprocal social interaction and impaired communication skills with restricted, repetitive, stereotypical interests and behaviors as specified by the *Diagnostic and Statistical Manual of Mental Disorders*, Fifth edition (DSM-5) ([American Psychiatric Association, 2013](#)).

Before 2013, **pervasive developmental delay (PDD)** and **Asperger syndrome** were considered to be related but separate disorders ([American Psychiatric Association, 2013](#)). Currently these disorders are included as part of the diagnosis of ASD as defined by the DSM-5. Rett syndrome is also a related developmental disorder but is not part of the ASD diagnosis because it is significantly different in causation, is progressive, and is almost exclusively found in girls ([Leonard et al, 2013](#)). The CDC estimates 1 in 59 children were identified with ASD in 2014, an increase of 15% from 2012. ASD is 4 times more common in boys than girls. Most children are diagnosed at 4 years of age, yet it can be reliably diagnosed earlier. Because early diagnosis is essential for best outcomes, the AAP recommends screening all children for autism at the 18- and 24-month well child visits.

Pathophysiology

The etiology of ASD is as complex as the individuals with the disorder. Many theories have been proposed in both the scientific literature and popular press, with some unsubstantiated claims causing harm, such as the discredited link between vaccines and ASD leading to a worldwide resurgence of measles. It is estimated that specific genes or gene mutations account for 10% to 25% of ASD cases, with interaction between genetics and environment assumed to be the primary cause ([Ornoy et al, 2016](#)). Various causal theories are listed in [Table 45.4](#).

ASD frequently occurs with other developmental or medical diagnoses; these are described in [Table 45.4](#). Children with ASD are more likely to have GI disorders than typically developing children, with up to 90% of children with ASD having some type of GI issue. Compared with children with developmental disabilities, constipation, bloating, or diarrhea are 3 times more common ([Chaidez et al, 2014](#)). GI problems are often expressed atypically as poor sleep, anxiety, aggression, or social withdrawal, resulting in lower quality of life ([Garcia et al,](#)

TABLE 45.4 Autism Spectrum Disorder: Proposed Causes, Contributors, and Associated Disorders

Causes of ASD	Disorders Associated With ASD
Genetics Presence of specific gene Gene mutation Gene deletion	Prematurity, small for gestational age (SGA)
Prenatal environmental exposure Pesticides Air pollution, particularly particulates Phthalates, BPA, solvent, flame retardant Medication: valproic acid, misoprostol, thalidomide Virus: rubella, CMV	Birth defects: present in 11% of children with ASD compared with 6.4% of typically developing children
Parental characteristics Older age of mother or father Maternal autoimmune disorder Parental mental health diagnosis Obesity	Genetic syndromes: 10% of all children with ASD have a genetic syndrome such as Down syndrome, fragile X, tuberous sclerosis, neurofibrosis, Angelman syndrome
Prenatal vitamin deficiency; folate	Psychiatric diagnosis: ADHD, anxiety disorders, depression
Birth spacing: conceived less than 18 months or more than 60 months	IDD: 30% with IDD, additional 23% borderline IQ Sleep disorders: present in two-thirds of individuals with ASD, 80% with presence of IDD and ASD Gastrointestinal disorders prevalence studies vary from 7% up to 90%, constipation and diarrhea primary disorders Allergies: primarily food allergies Seizure disorders with prevalence increasing with age Autoimmune disorders

ADHD, Attention-deficit/hyperactivity disorder; ASD, autism spectrum disorder; BPA, bisphenol-A; CMV, cytomegalovirus; IQ, intelligent quotient; IDD, intellectual and developmental disability.

References: Ornoy et al, 2016; Chaidez et al, 2014; Garcia et al, 2017; Campbell et al, 2014; Marshall et al, 2014.

2017). Reasons for GI problems include alterations in the gut microbiome (lack of diversity and characteristic microbiota), carbohydrate intolerance, and limited diets (Chaidez et al, 2014). Children with ASD frequently have restricted diets due to higher rates of food allergies, parent-imposed restricted diet, feeding problems, or selective “picky” eating (Marshall et al, 2014).

Overall, children with ASD who consume adequate nutrition grow well, although there is a subgroup of children who demonstrate a general overgrowth pattern with macrocephaly (Campbell et al, 2014).

Nutrition Assessment

Anthropometric measures. Height and weight are determined using standard equipment and growth charts. Head circumference

should be monitored up to 36 months, or longer if abnormal growth has been noted (Campbell et al, 2014). Weight gain and growth may be decreased with use of stimulant medicine, and excessive weight gain can occur with psychoactive medications such as risperidone (Ptomey and Wittenbrook, 2015; Richardson et al, 2017).

Biochemical measures. There is no standard pattern of tests other than regular newborn screening and blood work for health monitoring. If pica is noted or restrictive diets followed, pertinent nutrition blood work should be ordered. Because autoimmune disorders and allergies are more common, appropriate testing should be done when suspected (Ly et al, 2017). When psychotropic medicines are used, blood lipids, hemoglobin A1C, and liver enzymes should be followed due to risk of metabolic syndrome.

Dietary intake. Dietary assessment is important due to limited diets and frequent constipation (see Chapter 4). Up to 89% of children with ASD exhibit some form of eating issue compared with 25% of typically developing children (Marshall et al, 2014). Limited dietary intake coupled with rigid, monotonous eating patterns places children with ASD at risk for both nutrient deficiencies and excesses (Marshall et al, 2014; Stewart et al, 2015). Idiosyncratic dietary patterns such as avoiding a certain color of food, eating specific foods in specific locations, or avoiding certain food textures are common. One multicenter study found that up to 40% of children with ASD had probable nutrient deficiencies (Stewart et al, 2015). Fruits and vegetables are commonly low in the diet, while processed carbohydrates are high (Garcia et al, 2017). When variety is low and only a few foods are regularly consumed, the diet can be unbalanced, with high levels of sodium, poor-quality fat choices, and low fiber, especially if fortified foods or highly processed foods are consumed in large amounts. Pica or mouthing/chewing nonfood items can be a concern in ASD and can cause significant health problems, including hepatitis, parasitic disease, lead toxicity, and anemia. Including a feeding assessment along with a diet assessment (see Chapter 4) is helpful in determining whether feeding skills are age appropriate or if sensory processing is interfering with eating—both issues associated with ASD (Ptomey and Wittenbrook, 2015). Because social interaction is impaired, children often do not understand or display typical feeding cues and the feeding relationship may be challenged. Rigid eating habits, poor sense of hunger/satiety, textural aversions, and limited access to physical activity have resulted in a higher rate of obesity in ASD, especially among adolescents (Marshall et al, 2014).

Medical Nutrition Therapy

No one therapy or treatment modality works for all individuals with ASD (Marshall et al, 2014). Most children with ASD participate in a variety of interventions to address rigid or stereotypic behavior, sensory processing, and communication. These can include behavior intervention, structured educational approaches, medication, speech therapy, and occupational therapy. Interdisciplinary feeding therapy is warranted to address selective eating secondary to rigidity, motor planning, or sensory processing. Food rewards are commonly used in behavioral treatment, and alternatives should be explored because food rewards promote eating when not hungry, associate eating with reward, and are linked to emotional eating and obesity.

MNT is an effective complement to other treatments. Addressing selective eating habits can be challenging, and expanding the child's food repertoire takes time due to the rigidity inherent in the disorder. Treatment strategies need to respect the child's and parents' anxiety around mealtime because it can be great due to the child's rigidity, motor skills, and sensory preferences and parents' stress (Marshall et al, 2014). Common techniques include operant behavioral, systematic sensory desensitization, and food chaining (Marshall et al, 2014; Ptomey and Wittenbrook, 2015). Obesity, allergies, intolerances,

and GI complaints are all common in children with ASD yet often go untreated (Chaidez et al, 2014; Garcia et al, 2017; Ly et al, 2017). Obesity treatment can be especially challenging due to dietary rigidity, hypersensitivity to sensory characteristics of food, and limited access to physical activity. Specialized diets are often trialed in children with ASD both due to medical concerns (GI complaints, seizures, allergies) and media reports (Garcia et al, 2017).

The most popular diet has been the gluten-free (GF) casein-free (CF) diet. The majority of studies have not found measurable benefit, but parents often cite subtle improvements (Sathe et al, 2017). It has been theorized that the GF/CF diet or somewhat related specific carbohydrate diet (SCD) may be helpful in a small group of children because the diet is not treating the ASD per se but addressing undiagnosed immune or GI disorders, which are common in ASD (Garcia et al, 2017). Studies have documented the potential for nutritional deficits with the use of restrictive diets, specifically poor bone status with casein restriction, yet the risk can be minimized with dietitian supervision (Ly et al, 2017; Stewart et al, 2015). Tables 45.5 and 45.6 describe nutritional treatments commonly used in ASD. Care should be taken when beginning a diet trial, because once a food is removed from a child's diet, it may be difficult to reintroduce that food if the child is a selective eater. Conversely, if special products need to be used, such as GF alternatives, acceptance must be ensured before starting the diet trial to ensure dietary adequacy. Length of the trial and outcome measures should be clearly identified and the trial continued or ended based on measured effectiveness. When instituting a diet plan, the overarching consideration is to ensure a health-promoting, developmentally appropriate diet that allows for social inclusion at home and in the community.

When MNT is used, care providers and the child, when possible, are key contributors to the process, identifying priorities, preferences, and potential challenges (Garcia et al, 2017). Collaborating with occupational therapists, speech and language pathologists, and others on that child's team is critical for success, especially if a child is a highly selective eater. As a result of years of rigid and monotonous diets, use of food rewards, psychotropic drug use, and limited physical activity, rates of diet-related disorders such as hypertension, obesity, and diabetes are 40% to 50% higher in adults with ASD compared with the general population (Garcia et al, 2017; Marshall et al, 2014). Nutritional guidance throughout the life course is essential, directed not only toward the caregiver but also toward the adolescent with ASD, especially as they achieve independence (Garcia et al, 2017).

Attention-Deficit/Hyperactivity Disorder

Attention-deficit/hyperactivity disorder (ADHD) is a neurobehavioral problem beginning in childhood and often extending into adulthood. Between 9.5% and 11% of children in the United States have been diagnosed with ADHD (CDC, 2018). Nearly two-thirds of children with ADHD have another behavioral diagnosis. Diagnosis is based on the presence of specific behaviors noted in different settings: home, school, and community. According to the DSM-5 criteria for diagnosis, the child must consistently show signs of inattention, hyperactivity, and impulsivity inappropriate for age that interfere with daily functioning and are not attributed to other diagnoses. Based on symptoms noted, three types are differentiated: (1) predominately inattentive type, (2) predominately hyperactive-impulse type, and (3) combined presentation. Presentation may change over time as

TABLE 45.5 Proposed Dietary Interventions in Neurobehavioral Disorders

Dietary Interventions	Associated Disorders	Guidance for Use
Additive/food dye–free diet Removal of synthetic food dyes: Blue 1 & 2, Citrus Red 2, Green 3, Red 40, Yellow 5 & 6, sodium benzoate	May reduce hyperactive and attentional issues in ADHD, ASD, and typically developing children with sensitivities (Garcia et al, 2017; Lange et al, 2017; Ly et al, 2017)	Requires instruction in label reading, may increase cost of some food products
Elimination diets Elimination of common allergens, usually milk, egg, wheat, soy, peanuts, tree nuts, fish/shellfish May also exclude other foods limiting diet to a handful of hypoallergenic foods	Food allergies Eosinophilic esophagitis ADHD ASD	Used temporarily for identification of problematic foods Requires detailed logs of intake and behavior to detect patterns Foods introduced systematically Removal of food for prolonged time can lead to rebound allergy response in sensitive children and can lead to food refusal in ASD When foods are reintroduced, identify foods that the whole family can eat to encourage social inclusion Requires instruction in label reading, may increase constipation especially if highly processed grains are used
FODMAPs diet SCD Elimination of classes of carbohydrate to address low levels of digestive enzymes or alterations in microbiome	Gastrointestinal concerns such as Crohn disease, ulcerative colitis, carbohydrate intolerance ASD with gastrointestinal symptoms	Requires detailed education for both consumer and RDN, both diets include staggered restriction/introduction which is helpful for identifying target foods and minimizes risk, specifically with SCD. SCD can lead to nutrient deficits for young children with limited diets. SCD may limit variety of textures which may influence progression of textures in feeding therapy (see Appendix 28)
GF and CF diets Eliminate casein-containing foods (milk from cow, goat, sheep, etc.) and/or gluten-containing foods (wheat, rye, barley)	Celiac disease (GF), commonly used with DS, ASD, and ADHD Cerebral folate deficiency (CF) Food allergies	Identify foods that whole family can eat to encourage social inclusion, requires instruction in label reading, may increase constipation especially if highly processed GF grains are used. Ensure calcium adequacy with CF diet.

ADHD, Attention-deficit/hyperactivity disorder; ASD, autism spectrum disorder; CF, casein-free; DS, Down syndrome; FODMAPs, fermentable oligosaccharides, disaccharides, monosaccharides, and polyols; GF, gluten-free; RDN, registered dietitian nutritionist; SCD, specific carbohydrate diet. References: Sathe et al, 2017; Ly et al, 2017; Ptomey and Wittenbrook, 2015; Mastrangelo, 2018; Garcia et al, 2017.

TABLE 45.6 Examples of Effects of Nutrient Supplementation in Neurobehavioral Disabilities

Omega-3 fatty acid supplementation	ASD: Little evidence supports the effectiveness of omega-3 supplementation to improve core or associated ASD symptoms (Sathe et al, 2017) ADHD: Modest support for decreased lability, increased attention, and decreased oppositional behavior with GLA and EPA supplementation (Lange et al, 2017) FAS: Possible benefit prenatally and postnatally (Murawski et al, 2015)
Digestive enzymes	ASD: Evidence is inadequate to assess the effects of short-term digestive enzyme supplements (Sathe et al, 2017)
Vitamin B ₆	ALDH7A1 deficiency: Pyridoxamine 5'-phosphate oxidase deficiency Hyperprolinemia type II: Reduced seizure activity (Mastrangelo, 2018) ASD: Evidence inadequate to support improvements in function, used individually, in various forms or with magnesium (Garcia et al, 2017)
Folic acid, folacin	Cerebral folate deficiency: Possible reduced seizure activity, improved motor function (Mastrangelo, 2018) Mitochondrial disorders: Often included
B ₁₂ sublingual or injection	ASD: Few significant group communication or behavioral benefits (Sathe et al, 2017)
Multivitamin supplement: riboflavin, alpha lipoic acid (ALA), coenzyme Q ₁₀ ; often with folinic acid and L-carnitine added when warranted	Mitochondrial disorders: Standard treatment to improve muscle function (Parikh et al, 2015) ASD: May warrant trial for children with ASD if mitochondrial disorder is suspected
Magnesium	ASD: No evidence to support use (Garcia et al, 2017) ADHD: Evidence inadequate to support use (Lange et al, 2017)
Probiotics	ASD: No evidence to support use for ASD, but moderate evidence to support use in treating gastrointestinal issues such as constipation (Garcia et al, 2017) Other disorders: May help to relieve constipation

ADHD, Attention-deficit/hyperactivity disorder; ASD, autism spectrum disorder; EPA, eicosapentaenoic acid; FAS, fetal alcohol syndrome; GLA, gamma-linolenic acid.

the demands on the child increase. The causes of ADHD are not well understood, but genetic factors play a role. In addition, other potential causes or risk factors include (Lange et al, 2017):

- Alcohol, drug, or tobacco use during pregnancy
- Premature delivery or low birth weight
- Environment, exposure at a young age; lead, pesticides, institutional care
- Brain injury

Television, parenting styles, sugar, and/or family dysfunction do not cause ADHD, although they can exacerbate symptoms.

Nutrition Assessment

Anthropometric measures. Measurements of height and weight should be taken and recorded on a regular basis. Medications used in treatment may cause anorexia resulting in inadequate energy intake and slowing of growth, especially with long-term use (Richardson et al, 2017).

Biochemical measures. Follow recommendations for well child care with increased frequency if child is not gaining weight or has limited intake.

Dietary intake. A detailed dietary history should be obtained (see Chapter 4), especially in children who are exhibiting limited dietary intake. Evaluate the mealtime environment to identify ways to reduce distractions through changing seating or limiting electronics.

Medical Nutrition Therapy

Standard treatment includes behavioral therapy (47% of children) and/or stimulant medication (62% of children). Stimulants tend to reduce appetite and weight gain, especially in the first year of use (Richardson et al, 2017). This contributes to a small but statistically significant lower adult height with long-term treatment with potential negative effects on bone mineralization (Richardson et al, 2017). Because the ability to

feel hunger has been reduced, the child cannot be expected to appropriately modulate his or her own intake. Treatment should consider how to maximize interest in eating, first by adjusting medication timing or dosing when possible (i.e., eating breakfast before medication) (Richardson et al, 2017). When weight gain is a concern, care providers often resort to forcing, bribing, or offering higher fat/sugar foods to encourage intake. These practices rarely are successful and may contribute to the higher rates of obesity seen in adolescents and adults with ADHD. Encourage weight gain by offering calorically dense foods before medication dosing or when effect is waning; small and short frequent meals; and limiting visual/auditory distractions at meals (TV, tablets). In severe cases, a child may need extra support in school meal settings due to the amount of stimulation.

Restricted diets for ADHD have been promoted since the early 1970s, starting with the Feingold diet. Despite mixed evidence of success, many families chose to follow this diet, which eliminates naturally occurring salicylates and artificial food dyes (FD&C numbered dyes) and additives (Ly et al, 2017). Recent research suggests that dietary manipulation may be beneficial in a subset of children, specifically the use of restricted elimination diets (REDs) and elimination of artificial food dyes (Lange et al, 2017).

Studies on RED and food dyes suggest that there are children with ADHD who have greater sensitivity to food additives and may also have atypical allergic responses (Ly et al, 2017). Although there are higher rates of food allergies seen in ADHD, positive response to elimination of a food did not correlate with results of allergy testing, suggesting a nonimmunoglobulin E (non-IgE) response (Ly et al, 2017). Table 45.5 details dietary interventions commonly used.

Supplemental omega-3 fatty acids have also shown mixed results in children with both normal and low blood levels. Meta-analysis has suggested that there is a modest benefit for some children, dependent on age or "critical period," presence of comorbidities, and length of time of

use (Lange et al, 2017). Gamma-linolenic acid (GLA) and eicosapentaenoic acid (EPA) appear to be the most effective of the EFAs trialed (Lange et al, 2017). Studies with strict inclusion criteria indicate that there is a subgroup of children who do respond positively to supplementation with slightly improved emotional ability, increased attention, and decreased oppositional behavior (Lange et al, 2017). Typical “Western” diets are low in omega-3 fats and have been linked to ADHD, raising the question if supplementation alone is beneficial or if changing the diet to improve variety of fat intake and reduce food additive intake would have the same benefit as supplementation (Ly et al, 2017). Table 45.6 details use of nutritional supplements.

While research on diet and ADHD remains inconclusive and there is significant individual variation in response, reducing intake of processed foods, increasing intake of omega-3 fatty acids, and ensuring appropriate weight gain and growth should be part of MNT for ADHD.

Cleft Lip and Palate

Cleft lip and/or cleft palate (CL/CP), also known as orofacial clefts, are some of the most commonly occurring birth defects, with an incidence of 1 in 500 to 700 births worldwide (Shkoukani et al, 2013). There is a wide variation of incidence by ethnic origin and geographic location, with the highest and lowest incidence, respectively, in Japan and in South Africa. Cleft lip is a condition that creates an opening of the upper lip. It can range from a slight notch to complete separation in one or both sides of the lips and extending upward. If it occurs on one side of the lip, it is called a unilateral cleft lip; if it occurs on both sides, it is called a bilateral cleft lip. The cleft palate occurs when the roof of the mouth has not joined completely; it can be either unilateral or bilateral. Cleft palate can range from just an opening at the back of the soft palate or separation of the roof of the mouth with both soft and hard palate involved. CL/CP result from incomplete merging and fusion of embryonic processes during formation of the face. There is also a condition called submucous cleft palate in which there is incomplete fusion of the muscular layers of the soft palate with fusion of the overlying mucosa (Figs. 45.7 and 45.8).

Lip and palate development occur between 5 and 12 weeks’ of gestation. Lip development begins first, usually at 5 weeks’ of gestation, followed by the development of the maxilla prominences and the primary palate. Fusion of the hard palate is completed by 10 weeks’ of gestation and the soft palate by 12 weeks. Cleft lip, but not cleft palate, can sometimes be identified in utero with fetal ultrasound. CL/CP have multiple causes: genetic, environmental, and idiopathic. It is estimated that 20% of orofacial clefts are associated with underlying syndromes (syndromic clefts) such as 22q11.2-related disorders, Treacher Collins syndrome, and Stickler syndrome. Syndromic clefts are more likely to

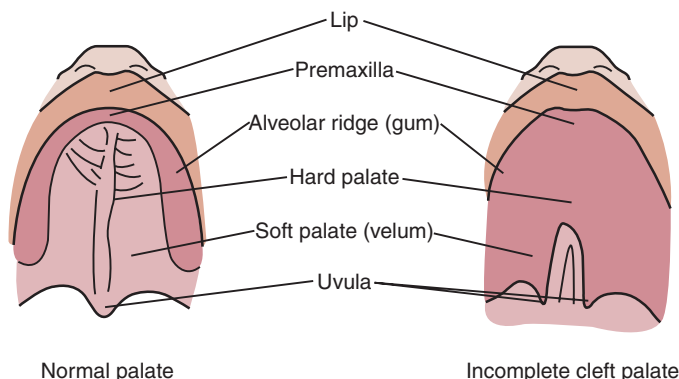


Fig. 45.7 Cleft palate. (From American Cleft Palate-Craniofacial Association, ACPA Family Services. Available from <http://acpa-cpf.org/families>.)

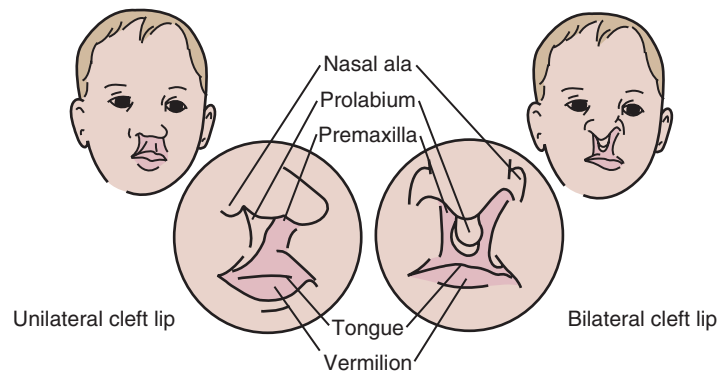


Fig. 45.8 Cleft lip. (From American Cleft Palate-Craniofacial Association, ACPA Family Services. Available from <http://acpa-cpf.org/families>.)

be genetic. Pierre Robin sequence (RS) is often referred to as a syndrome, but it is actually a set of abnormalities affecting the head and face consisting of a small lower jaw (micrognathia), a tongue that is placed further back than normal (glossoptosis), and breathing problems. A horseshoe-shaped cleft palate may or may not be present. Genetic counseling can now identify high-risk families for syndromic and nonsyndromic clefts (Leslie and Marazita, 2013). Environmental causes include teratogens such as maternal smoking, alcohol use, and use of antiepileptic medications.

Nutrition Assessment

Nutrition assessment for CL/CP includes the usual anthropometric measures for all infants and children. Biochemical measures are also those used with nonaffected children, and dietary intake information depends on the feeding problems that exist. Other problems include dental abnormalities and missing teeth, speech difficulties, and increased incidence of middle ear infections. Some infants (i.e., those with RS) may have increased difficulty breathing. The feeding evaluation is a major part of the assessment and is best accomplished with a team approach, including the parents. Because the major nutrition problem in CL/CP is feeding and providing adequate intake, growth can be jeopardized and must be assessed regularly.

Medical Nutrition Therapy

Treatment for CL/CP includes surgical treatment and nonsurgical treatment. The goal of surgical treatment is to repair the defect for good cosmetic and functional results. Nonsurgical treatment addresses how

TABLE 45.7 Feeding and Nutrition Goals for Children With Orofacial Clefts

- Effective feeding method with expected number of feedings and length of feeding in normal range of time
- Expected weight gain and growth for age, taking into account any related syndrome
- Feeding development typical for age (introduction of complementary foods at 6 months, introduction of cup drinking at 6 months)
- Maximize healing and minimize weight loss postsurgeries
- Healthy nutrition and oral health habits appropriate for age, including principles of Ellyn Satter’s Division of Responsibility in Eating
- For tube-fed infants and children, feeding plan that promotes oral intake, if appropriate

(Adapted from Lanier C, Wolf L: Children with cleft lip and/or palate: feeding and nutrition, *Nutrition Focus for Children with Special Health Care Needs* 32:6, 2017.)

TABLE 45.8 Surgical, Feeding, and Nutrition Management of Cleft Lip and Palate*

Age	Surgical Management	Feeding Modality	Nutrition Management
3–6 months	Lip repair	Cleft lip only: Breastfeeding or bottle feeding with wide-based nipple. Cleft palate: Specialty bottle. Upright position helps keep milk out of nasopharynx. Side lying position if tongue-based obstruction (Pierre Robin). Burp often secondary to increased air swallowing. After lip repair: Resume typical feeding modality soon after lip repair.	Breastmilk or standard infant formula. 400 IU vitamin D for babies receiving breastmilk. Introduction of complimentary foods at 6 months or when baby is showing readiness. Progression of textures typical for age. After lip repair: Resume breastmilk or formula,
9–15 months	Palate repair (syndromic clefts at late end) Ear tubes for otitis media	See above. Introduction of cup typical for age. Cup must not contain valve. Wean from bottle to cup typical for age (~age 1) or after palate repair. After palate repair: For 2–4 weeks, bottle feeding with specialty bottle and/or may use open cup or unvalved cup without hard or long spouts; nothing hard (no utensils or straws or fingers/hands) in mouth.	Wean to whole milk typical for age (age 1) but may wait until after palate is repaired. Toddler diet typical for age. Ensure meeting calcium and vitamin D requirement. After palate repair: For 2–4 weeks, soft, no chew diet.
2–5 years	Velopharyngeal insufficiency (VPI) surgery ("speech surgery") Nose and lip revision considered	After VPI surgery: Nothing hard (no utensils or straws) in mouth for 4–6 weeks.	Preschool diet typical for age. After VPI surgery: Soft, no chew diet for 4–6 weeks.
6–11 years	Bone graft of alveolar cleft after primary dentition complete and closure of oro-nasal fistulae Orthodontic interventions and teeth extractions as needed	After bone grafting: Nothing hard (no utensils or straws) in mouth for 4–6 weeks.	Diet typical for age. After bone grafting: Soft, no chew diet for 4–6 weeks.
12–21 years	Nose revision (rhinoplasty) if needed Jaw surgery in some cases (orthognathics) Orthodontic bridges and implants	After jaw surgery: Straws are ok.	Diet typical for age. After jaw surgery: Soft, no chew diet for 6–8 weeks; may need blenderized liquid diet.

*Timing of surgeries, feeding modalities, and nutrition interventions may vary between craniofacial centers.

(Adapted from Bernstein M, McMahon-Jones K: *Nutrition across life stages*. Burlington, MA, 2017, Jones & Bartlett Learning. Available from www.jblearning.com.)

to feed the infant until the cleft is repaired and how to optimize growth and nutrition and promote successful surgical treatment (Lanier and Wolf, 2017) (Table 45.7).

Surgical repair of the cleft lip generally is performed at 3 to 6 months of age, and cleft palate repair at 9 to 15 months. Other operations performed before the child starts school may include ear tubes for otitis media (usually performed at the same time as the palate repair), velopharyngeal insufficiency (VPI) surgery, and minor improvements to the lip or nose. Surgical interventions in later years include alveolar bone graft and, if needed, orthognathic (jaw) surgery (Table 45.8).

Breastfeeding is difficult for these infants because of problems with sucking, although those infants with just the cleft lip may be successful. It is generally recommended that the mother who wishes to breastfeed express her milk and give it to her baby from a specialized bottle and nipple. The use of an appropriate hospital-grade breast pump is recommended for milk expression. Parents and caregivers must be educated in the positioning of the child for feeding, nipple selection, bottle selection, feeding technique, and monitoring of intake (American Cleft Palate-Craniofacial Association, 2018).

Energy needs are generally the same as for a nonaffected infant. The exceptions to this are if the feeding process is too difficult (either adequate volumes are not achieved and/or energy needs are increased with the work of feeding) and if the infant has increased difficulty breathing (which results in increased energy demand). Strategies for solving these problems vary and include adjustment of the feeding modality (appropriate bottles and nipples) and feeding technique/infant positioning, and/or using more concentrated formula or breastmilk (Table 45.9 and see Chapter 43). Enteral support (tube feeding) is rarely needed, and this may be only for a short while.

Effective feeding requires that the infant be able to form a vacuum inside the mouth and form a seal around the nipple with the lips. This is achieved through the proper bottle, nipple, and position for feeding. Acceptable nipples and bottles include the Mead Johnson Cleft Palate Nurser, the Medela Special Needs Feeder, the Pigeon Bottle and Nipple, and the Dr. Brown's Specialty Feeding System (Dr. Brown's bottle with one-way feeding valve) (Table 45.10). Babies with CL/CP have unique feeding challenges; thus it is extremely important that an infant-feeding therapist or nurse experienced in feeding infants with clefts

TABLE 45.9 Increasing Calories Through Concentration of Formulas and Addition of Oils and Carbohydrates

Using 20 cal/oz Formula Caloric Density Required per Ounce		
Density Required per Ounce	Measures of Powder	Water Added
20 cal	1 scoop	2 fl oz
22 cal	2 scoops	3.5 fl oz
24 cal	3 scoops	5 fl oz
27 cal	3 scoops	4.25 fl oz
Using 22 cal/oz Formula		
22 cal	1 scoop	2 fl oz
24 cal	3 scoops	5.5 fl oz
27 cal	5 scoops	8 fl oz
Adding Oil or Carbohydrates Product	kcal	Source
Corn oil or safflower oil	9/g or 8.3/mL	Corn or safflower oil
Micro lipid	4.5/mL	Safflower oil
Medium chain triglyceride (MCT) oil	8.3/g or 7.6/mL	Fractionated coconut oil
Karo syrup	1 Tbsp = 58 kcal	Polysaccharides
Polycose liquid	2/mL or 60/fl oz	Glucose polymers
Polycose powder	3.8/g; 8/tsp; 23/Tbsp	Glucose polymers
Moducal powder	30/Tbsp	Glucose polymers

evaluate various types of equipment and carefully educate the parent in their use. Positioning in an upright position, choosing the appropriate bottle and nipple, and directing the liquid flow to the side or back of the mouth are strategies for optimal feeding. Babies with clefts swallow more air when feeding and should be given ample opportunities for frequent burping in an upright position.

Introduction of solid foods for the CL/CP infant can follow the usual protocol at 6 months of age or when the baby is showing readiness to feed (i.e., good head control and trunk stability and showing interest in food). Care that the food is presented slowly is important, allowing the infant to control each bite while gradually learning how to direct the food around the cleft. After the repair and healing of the cleft palate, feeding along the developmental pathway should progress slowly but normally (see [Chapter 15](#) and [Table 45.9](#)).

FETAL ALCOHOL SYNDROME

Fetal alcohol spectrum disorder (FASD) is a lifelong consequence of maternal alcohol consumption leading to growth retardation, facial malformations, and central nervous system impairment. Since identified in 1973 by Dr. Kenneth Jones and Dr. David Smith at the University of Washington School of Medicine, FASD has been documented throughout the world, with prevalence ranging from 2.4% to 20.8%; in high-risk populations such as Russian orphanages, rates have been estimated as high as 40% to 68% ([Jones et al, 1973](#); [Hoyme et al, 2016](#)).

Alcohol consumption during pregnancy interferes with normal fetal growth and development of all body systems and organs. Maternal nutritional status, use of multiple drugs, genetic susceptibility, and alcohol use pattern can influence the severity of the disorder ([Murawski et al, 2015](#)). Diagnosis of fetal alcohol syndrome (FAS) requires that the following criteria are fully met: (1) prenatal or postnatal growth deficiency, (2) FAS facial features, (3) abnormal brain structures and/or microcephaly, and (4) neurobehavioral impairment,

TABLE 45.10 Specialty Bottles

Specialty Bottle	How It Works	How to Pay and Where to Find
Mead Johnson Cleft Palate Nurser (Squeeze Bottle)	Milk is squeezed into the infant's mouth; flow is controlled by the person feeding.	Some insurance will cover Available through home care companies May be available at no cost from craniofacial centers Available to purchase on the Internet
Medela Special Needs Feeder (Haberman bottle)	The nipple has a large chamber with a one-way valve. Milk flows into the nipple chamber but cannot flow back into the bottle. Milk is extracted either by squeezing the nipple (by person feeding) or clamping down on the nipple (the baby controls the flow). The nipple has different sized cuts allowing for fast or slow flow.	Some insurance will cover Available through home care companies May be available at no cost from craniofacial centers Available to purchase on the Internet
Pigeon Cleft Palate Nurser	The nipple has a one-way valve and the rubber on one side of the nipple is very thin. Milk flows into the nipple and the baby can extract by clamping down on nipple: the flow is controlled by the baby.	Some insurance will cover Available through home care companies May be available at no cost from craniofacial centers Available to purchase on the Internet
^a Dr. Brown's Specialty Feeding System (Dr. Brown's bottle with one-way feeding valve) Nipple options include ultra preemie, preemie, and #1–4	Same feeding mechanism as Pigeon Nurser.	Not covered by insurance Bottles and/or valves may be available at no cost from craniofacial centers Available to purchase on the Internet

^aThe valve fits in the nipple of any Dr. Brown bottle.

(Adapted from Lanier, C, Wolf, L: Children with cleft lip and/or palate: feeding and nutrition. *Nutrition Focus for Children with Special Health Care Needs* 32:6, 2017.)

with developmental or cognitive impairment depending on age. The three facial features include smooth filtrum (the groove between the upper lip and the nose), thin upper lip, and small palpebral (between the upper and lower eyelid) fissures. Partial fetal alcohol syndrome (PFAS) is diagnosed when two of the criteria are met with known alcohol exposure, or three of the four criteria when exposure is not documented. **Alcohol-related neurodevelopmental disorders (ARNDs)** and **alcohol-related birth defects (ARBDs)** are additional diagnoses used to identify children without facial manifestations or poor growth but with alcohol-associated psychological or medical problems (Hoyme et al, 2016). Children with ARNDs may have intellectual disabilities and problems with behavior and learning, particularly with math, memory, attention, judgment, and poor impulse control. Children with ARBDs may have problems with the heart, kidney, bones, or hearing. Due to the clinical complexity, a multidisciplinary team should make the diagnosis (Hoyme et al, 2016).

Nutrition Assessment

Anthropometric measures are very important in the assessment of the FASD child due to characteristic growth deficiency. Prenatal growth restriction of the infant often persists postnatally despite a high caloric intake. Growth should be evaluated frequently and plotted on the CDC and WHO growth curves (Hoyme et al, 2016). Feeding problems are associated with FAS, including oral-motor problems expressing as poor suck in infancy and delayed feeding progression in toddlers. Poor growth and feeding issues often result in the diagnosis of failure to thrive (Amos-Kroohs et al, 2016). Impulse control, food hoarding, and hyperphagia are common in older children (Amos-Kroohs et al, 2016; Ptoimey and Wittenbrook, 2015).

Medical Nutrition Therapy

Use of MNT with FASD emphasizes prevention of secondary disability beginning with prenatal vitamin-mineral supplementation when alcohol consumption is suspected because this may reduce disability both by addressing maternal nutritional deficits and mitigating teratogenic effects of alcohol (Murawski et al, 2015; Popova et al, 2016). While breastfeeding generally is encouraged, excessive alcohol use is a contraindication. Considering the array of potential nutritional concerns, MNT for the child with FASD is focused on the specific nutrition problem that exists for that child. Motor impairment, ADHD, developmental delay, seizure disorders, communication challenges, hyperphagia, and sensory impairments (visual/hearing) are common comorbidities that can put nutrition at risk (Amos-Kroohs et al, 2016; Popova et al, 2016). Short stature, sensory processing issues, and impulsivity increase the risk of overweight or obesity in the older child and adolescent. In addition, parental substance abuse, past or present, may impact the parent-child feeding relationship, requiring supportive intervention with referral to social services as needed (Amos-Kroohs et al, 2016).

Overall, caloric and nutrient needs do not differ from an unaffected child, although poor fetal nutrition or feeding problems may increase the risk of deficiencies (Popova et al, 2016). Vitamin, mineral, omega fatty acid, and choline supplementation may be warranted to maximize brain development and address prenatal deficits (Murawski et al, 2015). It can be difficult to determine caloric need in the young child when growth restriction is present. Increasing calories potentially can support improved growth but may also lead to excessive weight gain without increasing the rate of growth. Comparing serial measurements of weight to linear gains is necessary to individualize calorie recommendations. Strategies for increasing calorically dense foods may be required for those who have difficulty gaining weight. Food texture may need adaptation if oral-motor impairments are present.

FASD is currently the leading preventable developmental disability in the world. Education regarding FASD should occur in all women of childbearing age because there is no safe time or amount of alcohol during pregnancy (Murawski et al, 2015).

Complementary and Integrative Medicine in Intellectual and Developmental Disabilities/Autism Spectrum Disorders

Current standard treatments for IDD/ASDs include therapies (occupational, physical, speech/language), medication, and MNT. Often, standard interventions are time consuming, are expensive, or have negative side effects (Lindly et al, 2018). These treatments may reduce disease progression and improve function, but they may not provide significant reduction of the child's overall disability. In hoping for greater improvements, caregivers and health professionals seek out complementary and integrative medicine (CIM) to maximize development and quality of life. Over the past 50 years, a variety of studies have been initiated to assess if dietary manipulation or nutritional supplementation can treat or even cure developmental diagnoses. With the exception of the ketogenic diet for seizures and dietary management of metabolic disorders/inborn errors of metabolism, research on supplementation or dietary manipulation has found minimal or transitory improvements in developmental, behavioral, or functional measures (Lewanda et al, 2018; Mastrangelo, 2018; Sathe et al, 2017). As indicated in Chapter 11, research in CIM can be difficult to interpret in pediatrics and even more so in children with IDD/ASD, a very heterogeneous group likely to take multiple medications, participate in multiple therapies, and require frequent medical intervention—all of which will confound research results (Sathe et al, 2017). Studies may have bias, small sample size, multiple agents trialed in one study, lack of long-term follow-up, and flawed assessment of both nutritional adequacy and behavioral/functional change (Garcia et al, 2017; Lewanda et al, 2018; Brizee, 2019; Sathe et al, 2017). Harm is often underassessed, and the long-term effects of supplementation and dietary restriction are unknown (Garcia et al, 2017; Stewart et al, 2015). Despite the lack of clear recommendations, media, consumer groups, and supplement-industry sponsored conferences promote nutritional supplements and special diets in this population. This has resulted in greater use of CIM in this population compared with typically developing children (Lindly et al, 2018; Stewart et al, 2015).

Dietitians and caregivers are challenged with determining whether the potential for benefit from a diet or supplement outweighs the potential risk. Negative side effects include direct effects such as a nutrient deficit or excess and also indirect effects such as the financial and time costs. This can be considerable with use of multiple supplements, need for special dietary products, and payment for practitioners, many of whom are not covered by insurance (Lewanda et al, 2018; Lindly et al, 2018; Stewart et al, 2015). There is also the risk that, considering time and finances, unproven interventions will replace established, evidenced-based treatments (Lindly et al, 2018).

While various CIM methods, including herbs, yoga, and massage, are accessed in this population, the primary CIM modalities used are nutritional supplementation and dietary manipulation; see Chapter 11 for discussion of a wider range of CIM (Lindly et al, 2018).

Nutritional Supplements

Nutritional supplementation has shown benefits in rare metabolic disorders and mitochondrial disorders, but evidence that supplementation improves function in most developmental disorders is lacking (Lewanda et al, 2018; Mastrangelo, 2018; Parikh et al, 2015; Sathe et al, 2017). Other than in those examples, supplementation has been studied without clear or replicated success (supplements studied include B vitamins—B₆, B₁₂,

folic acid, biotin, and riboflavin—vitamin E, l-carnitine, magnesium, zinc, and iron) (Garcia et al, 2017; Lewanda et al, 2018). While there often are changes in biochemical indices, significant parallel changes in behavior or function have not been found or replicated (Sathe et al, 2017). Meta-analyses and review articles caution on making supplement conclusions or recommendations, due to flawed research methodology. Although research methods are improving, evidence remains insufficient to recommend use of nutritional supplementation (Garcia et al, 2017; Lewanda et al, 2018; Ly et al, 2017; Sathe et al, 2017).

Omega fatty acids and probiotics both have moderate evidence for improvements in mood and attention yet are not necessarily specific to IDD/ASDs. It is unclear if this is a direct result of the supplement or of rectifying unbalanced dietary fats (Lange et al, 2017; Ly et al, 2017). Probiotics are thought to improve GI dysbiosis, immune function, and inflammation in ASD and potentially may have more global benefit for ASD and other disabilities (Chaidez et al, 2014; Garcia et al, 2017). This is an emerging area of research; use of probiotics has limited harm but may be contraindicated in children with immune suppression, and it may initially cause bloating and gas (Garcia et al, 2017).

Although data are tenuous, individuals with IDDs use supplements twice as frequently as the general population, with use highest when children are young (Lindly et al, 2018). A study of children enrolled in the Autism Treatment Network revealed that more than 50% of children with ASDs used nutritional supplements compared with 32% of typically developing children (Stewart et al, 2015). This same study found that supplementation did not address actual areas of deficit and led to excessive nutrient intake. This emphasizes the importance of RDN-supervised supplement use; initiating discussion when clients request use; addressing deficits indicated by poor diet or blood testing; and minimizing harm through attention to dosage, especially when fortified foods are consumed or contraindications exist such as calcium supplementation in Williams syndrome (Ptomey and Wittenbrook, 2015; Stewart et al, 2015). Table 45.6 summarizes use of supplementation. See Chapter 11 for more information about the safe recommendation of dietary supplements.

Special Diets

In 1954, the discovery that restriction of dietary phenylalanine in the diet of a child with phenylketonuria (PKU) prevented intellectual disability created hope that diet could treat or even cure developmental disabilities. Dietary treatments have been proven effective for some inborn errors of metabolism and for minimizing seizures but have not yet shown similar success in treating other disabilities (Mastrangelo, 2018; Ptomey and Wittenbrook, 2015). Although there is a lack of empirical evidence, parents report that they chose CIM because they were unhappy with traditional interventions and felt more comfortable and in control using food rather than medication as treatment (Lindly et al, 2018; Ly et al, 2017). Studies of the most popular diets—ketogenic diet, GF diet, CF diet, and SCD—have not found significant behavioral or functional change (other than the noted seizure control), yet caregivers chose to continue the diet after the study ended. It is suggested that parental perceptions of change may be influenced by desire for change, emotional investment in the treatment, and misinterpretation or attribution of any improvement to dietary change (Garcia et al, 2017; Lindly et al, 2018). Conversely, the subtle improvements noted may not be variables being measured, and therefore parents see benefit when researchers do not. For example, parents may see improved bowel movements or sleep, which are not assessed. Use of a special diet may be addressing underlying anemia, GI discomfort, or allergy, reducing discomfort and irritability (Garcia et al, 2017; Ly et al, 2017).

A nutrition assessment may reveal reasons to trial a diet such as a history of or current allergy, reflux, or constipation. The use of special diets

is not without nutritional risk, but this can be minimized with RDN supervision. In addition to the risk of nutrient deficits, especially when entire food groups are eliminated or severely restricted, prescribed diets with very limited variety can exacerbate existing conditions such as constipation through dependence on highly processed foods. Children with special needs are often socially isolated, and restricted diets can further isolate the child by making eating with others challenging. Table 45.5 describes diets frequently used in this population.

Choosing Complementary and Integrative Medicine Treatments

Children with neurodevelopmental disorders are at greater risk of harm from both conventional and CIM intervention. Drug-nutrient interactions, GI problems, or altered immune status may make negative response more likely while concurrently making use of CIM attractive for resolving outstanding issues. The RDN can facilitate effective use of CIM by helping parents and other health care providers better understand the applicability and veracity of research (Brizee, 2019; Garcia et al, 2017; Ptomey and Wittenbrook, 2015).

Products that are promoted or advertised as follows should be avoided:

- Personal anecdotes, celebrity endorsement, or industry-backed research are the only evidence for effectiveness.
- Claims of effectiveness are grandiose or universal and apply to many different disorders; the magnitude of improvement is great, and the product is claimed as a cure.
- Reported benefits are general, vague, and unmeasurable.
- Exclusive to this company with a “special” formulation that is a proprietary blend; company will not provide ingredients when asked (often necessary for safety concerns).
- Advertised as “opposed by the medical establishment.”
- Expensive or costly for what is offered, requires extensive testing that is only available through a partner site, or special promotions that require prepayment; unable to use insurance.
- Clay baths, enemas, and chelation therapy are promoted to remove toxic chemicals to cure or treat various conditions. These procedures can be harmful and have resulted in death. The FDA has issued a warning against use of chelation therapies and highlights the risk of lead exposure in clay baths.

Although there is risk for harm, this can be minimized with nutritional monitoring. In some cases the exploration of integrative therapies has led to successful treatment and a better understanding of brain, immune, and GI function in children with neurodevelopmental concerns. Each child should be assessed individually to determine whether any dietary or CIM therapy is appropriate.

COMMUNITY RESOURCES

For many types of nutrition problems and MNT, the school system is an excellent resource through the school lunch and school breakfast programs. Children and adolescents may receive modified meals at school. Child and adult care food programs must provide meals at no extra cost for children and adolescents with special needs and developmental disabilities. School food service is required to offer special meals at no additional cost to children whose disabilities restrict their diets as defined in the U.S. Department of Agriculture’s nondiscrimination regulations.

The term “child with a disability” under Part B of IDEA refers to a child evaluated in accordance with IDEA as having one of the 13 recognized disability categories: (1) autism; (2) deaf-blindness; (3) deafness; (4) mental retardation; (5) orthopedic impairments; (6) other health impairments caused by chronic or acute health problems such

as asthma, nephritis, diabetes, sickle cell anemia, a heart condition, epilepsy, rheumatic fever, hemophilia, leukemia, lead poisoning, or tuberculosis; (7) emotional disturbances; (8) specific learning disabilities; (9) speech or language impairment; (10) traumatic brain injury; (11) visual impairment; (12) multiple disabilities; and (13) developmental delays. Attention-deficit disorder may fall under one of the 13 categories.

When a referral is made to the school system for a special meal related to a developmental disability, it must be accompanied by a medical statement for a child with special dietary needs. The request

requires an identification of the medical or other special dietary condition, the food or foods to be omitted, and the food or choice of foods to be substituted. The statement requires the signature of the physician or recognized medical authority. The school food service may make food substitutions for individual children who do not have a disability but who are medically certified as having a special medical or dietary need. An example is the child with severe allergies or an inborn error of metabolism. The availability of school food service for children with developmental disabilities is an important resource in the long-term implementation of MNT.

CLINICAL CASE STUDY

Child with Down Syndrome

Nutrition Assessment

Client History

Colin is a 21-month-old boy with Down syndrome. He was born prematurely (30 weeks' of gestation) and was started on nasogastric tube feeding at 10 days of age because of his poor weight gain and severe gastroesophageal reflux. The poor weight gain was caused by a weak suck, although swallowing was not a problem. After hospital discharge he was first seen by a nutritionist in an early intervention program at 4 months of age when it was noted that he had a gastrostomy tube placed at age 2 months.

Food/Nutrition-Related History

Colin received tube feeding instructions while hospitalized for his G-tube placement. He participated in early intervention oral-motor therapy. He was seen every 3 to 6 months for nutrition assessment and adjustment of G-tube formula to meet nutritional needs.

At 16 months Colin was tube-fed PediaSure and just started eating table foods. His usual intake was 1 jar of baby food per day along with the tube-feeding formula.

Anthropometric Measurements

At 4 months of age Colin was 22.5 in long (<5th percentile) and weighed 10 lb 7 oz (<5th percentile). Weight/length was 10th to 25th percentile. At 21 months, height was 28 in (<5th percentile) weight was 18.5 lb (<5th percentile). Weight/length was 25th to 50th percentile.

Nutrition-Focused Physical Findings

Colin is crawling but not yet walking, and he has very limited self-feeding skills. Now at age 21 months, his mother's highest priority is to stop the tube feeding and have Colin continue to grow well. She is concerned about his rate of weight gain. She also is concerned that constipation has become a problem requiring use of the medication lactulose. He also has respiratory problems and extreme hypotonia.

Feeding problems identified:

- History of weak suck and swallow
- Hyperactive gag reflex
- Refusal or inability to drink formula from a bottle or cup
- Poor appetite for oral foods
- Not self-feeding

Nutrition Diagnostic Statements

- Self-feeding difficulty related to developmental delays and feeding difficulty as evidenced by poor weight gain and below 5% weight/age.
- Inadequate oral intake of food and fluids related to feeding difficulty as evidenced by need of supplement to meet nutritional needs by G-tube.

Nutrition Intervention

1. Tube-fed toddler with daily caloric requirement estimated at 670 (estimated energy requirement [EER] = ~80 cal/kg).
2. Work with occupational therapist (OT) to determine chewing and swallowing challenges and to support development of safe eating and drinking.
3. Introduction of oral foods and fluids as recommended by OT before gastrostomy feedings.
4. Reduction of tube feedings to support transition to oral eating and to meet estimated needs.

Nutrition Care Questions

1. What would be your approach in working with this mother and the other team members?
2. What do you think would be Colin's nutritional needs, starting with energy?
3. How many ounces of a 30 cal/oz tube-feeding formula would you recommend for Colin to promote weight gain?
4. What steps should be taken to increase Colin's oral intake and decrease the tube feeding?
5. What would you recommend for management of his constipation?

CLINICAL CASE STUDY

Adolescent with Autism Spectrum Disorder and Obesity

Nutrition Assessment

History

Alex, a 16-year-old boy with autism spectrum disorder (ASD), was admitted to a residential facility after being removed from his special education class. Alex has age-appropriate receptive language but expresses himself usually using phrases from superhero TV shows and movies. Three months before referral, he began a trial of risperidone to address increasing irritability and aggression both at home

and at school. Medical evaluation identified sleep apnea, elevated liver enzymes, hypertriglyceridemia, and impaired fasting blood glucose. The contributing factor to the medical problems is his severe obesity.

Food/Nutrition-Related History

Alex's birth weight was 7 lb 8 oz, his birth length was 19 in, and he was full term. He had typical feeding development until 2 years of age, when he gradually

CLINICAL CASE STUDY—cont'd

Adolescent with Autism Spectrum Disorder and Obesity

started refusing previously accepted foods. Gradually, his intake was reduced to primarily crunchy foods including chicken nuggets, fish sticks, French fries, green apples, celery, most crackers, any type of Cheerios, chips, bacon, and beef jerky. Portions are large. His parents have tried to introduce more fruits and have had some success with other crunchy green vegetables such as green peppers or raw broccoli and freeze-dried fruits. Alex drinks milk, usually 24 to 30 oz a day, and soda is provided as a reward. If he is denied food, he tends to throw a tantrum and family members often relent.

Alex has been overweight since 2 years of age, when his weight for age was between the 85th and 90th percentiles. When he began middle school, his rate of weight gain increased, with his body mass index (BMI) at the 95th percentile. Since starting high school, his rate of weight gain has continued to increase, which his mother attributes to the role of food in the classroom as part of living skills training and for rewards. Since starting the risperidone, he has gained an additional 15 lb in 3 months.

Alex enjoys physical activity when it can be related to superheroes. He enjoys watching and playing soccer but has limited opportunity.

Anthropometric Measurements:

Weight: 210 lb (95.4 kg), >97th percentile

Height: 66 in (167.6 cm), 25–50th percentile

BMI index: 34 kg/m², 120% to 130% of the 95th percentile, classified as severe obesity

Biochemical Data

Total cholesterol: 210 mg/dL (high)

High-density lipoprotein (HDL) cholesterol: 29 mg/dL (low)

Triglycerides: 580 mg/dL (very high)

Glucose: 120 mg/dL (high)

Hemoglobin A1C: 6.1% (high)

Liver enzymes: aspartate aminotransferase (AST) 74 U/L, alanine aminotransferase (ALT) 124 U/L (high)

Nutrition-Focused Physical Findings

Excessive appetite

Poor hygiene

Sedentary with excessive screen time

Constipation (2 to 3 bowel movements [BMs] per week)

Nutrition Diagnostic Statements

- Obesity, pediatric related to excessive food and beverage intake, large portion sizes, and high-calorie food choices, as evidenced by BMI of 34.
- Altered nutrition-related laboratory value related to obesity and risperidone use as evidenced by elevated cholesterol and triglycerides and hemoglobin A1C of 6.1%.
- Altered nutrition-related laboratory value related to obesity and excessive soda intake as evidenced by elevated liver enzymes ALT and AST.
- Altered gastrointestinal (GI) function related to low activity level and insufficient fiber and fluid intake as evidenced by report of only 2 to 3 BMs per week.

Nutrition Intervention Goals

1. Work with family and school to decrease portion sizes, especially of highly processed foods
2. Increase fiber and fluid intake
3. Reduce sugary beverages due to elevated triglycerides, A1C, and liver enzymes
4. Increase access to physical activity

Nutrition Care Questions

1. When asked to limit food at school, teachers report that reward tokens to purchase snacks at the student store are part of the living skills curriculum. What would be your response?
2. What tools can you offer Alex and his family to help manage portion size and increase fruit/vegetable intake?
3. What resources are available to help Alex increase his physical activity?
4. What suggestions can you make to help Alex's parents respond to his demands for food?

USEFUL WEBSITES

American Cleft Palate Craniofacial Association
 Association of Maternal and Child Health: Children and Youth with Special Healthcare Needs Program
 Centers for Disease Control and Prevention: Birth Defects Research March of Dimes
 National Autism Association
 National Center for Education in Maternal and Child Health
 National Dissemination Center for Children with Disabilities
 National Folic Acid Campaign
 National Human Genome Research Institute: Atlas of Human Malformation Syndromes in Diverse Populations
 NIH National Institute of Mental Health (ASD)

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